

PanelGaurd Robotic System

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Abstract

The aim of this project is to design and develop a semi-autonomous robot for the purpose of cleaning solar panels to improve their performance and efficiency. Over time, dust and other environmental factors can reduce the effectiveness of solar panels, leading to decreased energy production. By analyzing existing robotic cleaners, this project integrates optimized features for efficient panel cleaning. The robot uses a chassis made of aluminum sheets and manufactured using CNC laser cutting with a tank-like design, and the mechanical movement of the robot is achieved using timing belts, providing an efficient and reliable movement mechanism across the panel surface. A motor-driven circular brush connected to a yaw joint for horizontal sweeping, linked with an additional joint enabling precise positioning of the cleaning head. Infrared and ultrasonic sensors enable the robot to detect edges in addition to any unpredictable obstacles ensuring safe and efficient navigation. Motor sizing was successfully conducted ensuring sufficient torque and power for the cleaning tasks, considering the robot's weight, surface, and desired moving speed. The robot is powered by a rechargeable lithium battery and controlled wirelessly through a remote control, enhancing user-friendliness. By employing inexpensive materials and a simplified design, this project delivers a cost-effective and reliable solution for automated solar panel cleaning, reducing complexity for end users. The successful development of this robot contributes to improved energy generation efficiency, and the long-term sustainability of solar power systems.

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Chapter 1: Introduction

In the face of increasing energy demands and environmental concerns, the utilization of solar panels for renewable energy generation has become paramount. However, the efficiency of these panels can be significantly hampered by dust accumulation and other environmental factors [1]. This report introduces a project aimed at addressing this issue through the design and development of a semi-autonomous robot. The robot is specifically engineered for the cleaning of solar panels, thereby enhancing their performance and efficiency. PanelGaurd seeks to maximize renewable energy generation from solar panels. The following sections will provide a detailed definition of the project, including objectives, and constraints.

1.1 Detailed Definition of the Project

PanelGaurd focuses on the design, development, and manufacture of an economical remote-controlled automated system for cleaning solar panels. The project conducts research into the various robotic solar panel cleaners currently available, focusing on their mechanical movement on inclined surfaces and operational processes.

The robotic cleaner is constructed on an aluminum chassis, Control instructions for movement are wirelessly transmitted to the robot via a remote controller.

For mechanical locomotion, the robot employs gear wheels connected to rubber belts. This design ensures that the robot maintains proper adherence to the surface of the solar panels while distributing stress evenly across the surface.

The cleaning apparatus of the robot consists of circular motorized brushes positioned at the front of the system, sweeps and clean with water sprayers, these components work together to effectively remove accumulated particles such as dust and dirt from the surface of the solar panels.

This detailed project definition serves as a comprehensive guide to the design and development process, ensuring the project meets the desired objectives.

1.2 Objectives

The aim is to develop a low-cost solar panel cleaning system by using inexpensive materials and components as well as economical manufacturing processes.

Additionally, the robot integrates user friendly operations by reducing the complexity of control making it easy to use and operate for end users. To provide a much wider perspective of how the robotic solar panel cleaner has developed. Below is an outline of the main goals and objectives for developing the system.

- **Design for Cost:** This approach is aimed at minimizing the cost of developing the cleaning system without limiting the robot's performance or deviating from the main objectives.
- **Design for Manufacturability:** The robot is designed to make it easy to manufacture using available materials while also outsourcing related components and implementing economical manufacturing processes.
- **Design for Safety:** This project incorporates safety features to ensure the robot can be used and operated safely without causing harm to the user or the solar panels.
- **Enhancing Efficiency:** Improve solar panel efficiency by effectively removing dust and debris.
- **Adaptability:** The robot is designed to adapt to different solar panel installations, considering variations in height and inclination.
- **Robustness and Reliability:** Ensure that the robot can withstand outdoor conditions (temperature variations, humidity, and dust) and operate reliably over extended periods. Regular maintenance and fault tolerance mechanisms should be considered.

1.3 Benefits of PanelGaurd

- **Time-Saving Capabilities:** Once set up, a robot cleaner can autonomously navigate your space and clean your desired places.

Environmental Impact: Using a robot ensures that the cleaning process is environmentally friendly. It contains no chemical by-products and promotes sustainable energy production.

- **Remote and Wireless Operation:** The robot can be operated remotely, eliminating the need for manual labor. Workers aren't put in danger while cleaning solar panels located in challenging or elevated positions.

Solar panel cleaning robots eliminate the need for human workers to physically access the panels, significantly reducing the risk of accidents and injuries. This enhanced safety ensures a safer work environment for maintenance staff.

1.4 Cleaning robots Drawbacks

While solar panel cleaning robots offer several benefits, they also have certain drawbacks PanelGaurd tried to minimize it.

- **Complexity and Maintenance:** Robotic systems involve intricate components, sensors, and software. Regular maintenance, software updates, and occasional repairs are necessary to keep the robots operational.
- **Risk of Damage:** Despite efforts to design gentle cleaning mechanisms, there's always a risk of unintentional damage to the panels. A malfunctioning robot or abrasive cleaning materials could scratch or crack the glass surface.
- **Limited Mobility:** Having struggle in extreme panel orientation.
- **Limited Water Supply:** Solar panel cleaning robots need a constant water supply for effective cleaning. If the robot's water tank is too large, it can become heavy, potentially damaging the panels, or making them unable to support its weight.

Given the constraint of limited water supply and the potential issues with large onboard water tanks, an external water source linked to the robot via a hose offers a compelling solution.

1.5 Project Constraints

Without any limitations the project possesses the potential for exceptional quality and outstanding performance. However, the presence of constraints imposes certain limitations on the project's design and execution [2].

Therefore, engineering design must carefully consider numerous constraints, including economic, environmental, safety, sustainability, functionality, and manufacturability factors.

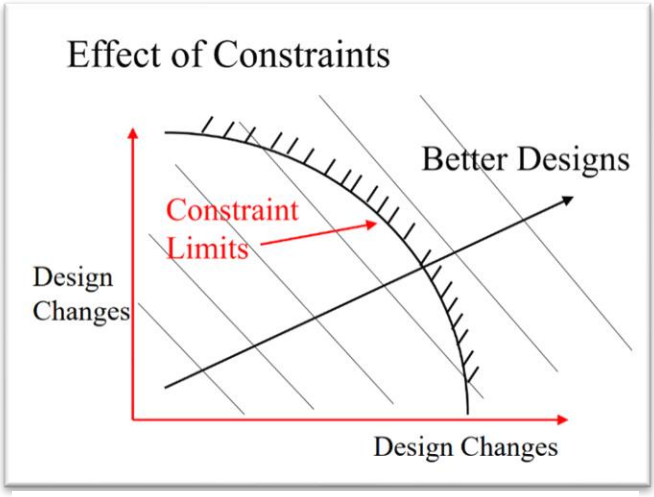


Figure 1 effect of constraints.

- **Economic constraints.** Cost is an important parameter to consider. As the project must be economically affordable, therefore efficient use of budget for the materials, components, and manufacturing is of importance.
- **Environmental constraints.** Components like batteries should be properly disposed of and also the plastic features of the robot can be recycled, and good control of waste disposal should be considered [3].

- **Safety constraints.** Implementing a user-friendly design which poses no risk of harm or damage to the user, or solar panels should be considered. This would mean that the structure of the cleaning robot should be devoid of sharp edges and include proper insulation of the electrical units to prevent issues like short circuits, overheating.
- **Sustainability.** In order to enhance the life span of the robot, it is imperative to implement certain precautionary measures. Such as water resistance and overheat protection.
- **Functional Constraints**. for optimal efficiency, solar panels should be oriented southward at an angle of 30°-45° [4]. To ensure stability and traction on the panel, the robot must have proper weight distribution and friction. This is why tracked wheels were selected to expand the contact area, ultimately enhancing both friction and balance.

Chapter 2: Literature Review and Theoretical Background

The world is increasingly leaning towards cleaner, non-conventional, and renewable energy sources [5]. Among these, the Sun stands out as an inexhaustible and abundant source of energy. Photovoltaic (PV) Solar Panels have enabled us to harness solar power and convert it into electrical energy with notable efficiency and convenience [5].

Before designing a cleaner device for solar panels, it is crucial to thoroughly understand the structure of the solar panel module, including its mechanical and electrical properties. Furthermore, an investigation into the module's construction and the technologies employed in the solar electrical system is essential.

2.1 Solar cell history and technology

2.1.1 Early History of Electricity Generation:

The first recorded observation of electricity dates to ancient Greece when Thales of Miletus observed that amber, when rubbed with fur, could attract light objects. This phenomenon, known as static electricity, laid the foundation for future discoveries.

In the 19th century, scientists like Alessandro Volta and Michael Faraday made significant breakthroughs in understanding and harnessing electricity. Volta invented the first chemical battery, while Faraday discovered the principles of electromagnetic induction, which led to the development of electric generators.

2.1.2 The Birth of Photovoltaics:

The photovoltaic effect, the conversion of light into electricity, was first observed by French physicist Alexandre Edmond Becquerel in 1839. He noticed that certain materials, when exposed to light, produced a small electric current [6]. However, it wasn't until the mid-20th century that practical solar cells were developed.

2.1.3 (PV) vs (CSP) Systems:

Photovoltaic (PV) and Concentrated Solar Power (CSP) systems are two distinct solar energy technologies. PV panels directly convert sunlight into electricity through the photovoltaic effect, while CSP systems use mirrors or lenses to concentrate sunlight onto a receiver, heating a fluid that drives a turbine to generate electricity. PV systems generally have lower initial costs and are more suitable for smaller-scale applications, while CSP systems offer higher efficiencies and the potential for energy storage through thermal energy storage systems, making them more suitable for large-scale power plants [7].

Solar energy technologies, encompassing both Photovoltaic (PV) systems and Concentrated Solar Power (CSP) systems, offer promising solutions for sustainable energy generation. While CSP systems utilize concentrated sunlight to heat a fluid and drive turbines, PV systems directly convert sunlight into electricity through the photovoltaic effect. Understanding the intricacies of PV cells, including their materials, design, and efficiency factors, is crucial for optimizing their performance and maximizing energy output. This knowledge is particularly valuable in the context of developing an integrated robotic system for cleaning PV modules, as it allows for a targeted approach to enhancing solar energy generation by mitigating the impact of dust, dirt, and other contaminants on cell efficiency, ultimately contributing to the broader goals of advancing solar energy technologies [8]. Table 3 provides a simplified overview of the solar energy technologies discussed above.

Table 1: Photovoltaic (PV) vs Concentrated Solar Power (CSP) Systems

Feature	Photovoltaic (PV)	(CSP)
Energy Conversion	Direct conversion of sunlight into electricity	Indirect conversion
Efficiency	(15-20) %	(20-30) %
Cost	Generally lower initial investment.	Higher initial investment
Land Use	Requires more land area for the same power output.	Requires less land area for the same power output
Energy Storage	Requires battery storage for nighttime operation. (off grid)	Can incorporate thermal energy storage for nighttime operation
Reliability	Affected by weather conditions	Less affected by weather conditions
Maintenance	Lower maintenance requirements	Higher maintenance requirements
Scalability	Can be scaled from small residential systems to large utility-scale plants	Primarily suitable for large-scale power plants
Environmental Impact	Minimal environmental impact	Potential environmental impact due to land use and water consumption

In this report, the primary focus will be on PV cells. This is because Concentrated Solar Power (CSP) systems typically require a deeper understanding of thermodynamics and necessitate a thorough knowledge of the properties of the working fluid (such as steam). Given that our project aims to develop an integrated robotic system for cleaning PV modules, a detailed exploration of PV cell technology is more relevant to the scope of this report.

2.2 Solar panels

Types of solar panel modules can be classified based on three categories: material composition and manufacturing process, cell size, and construction and design.

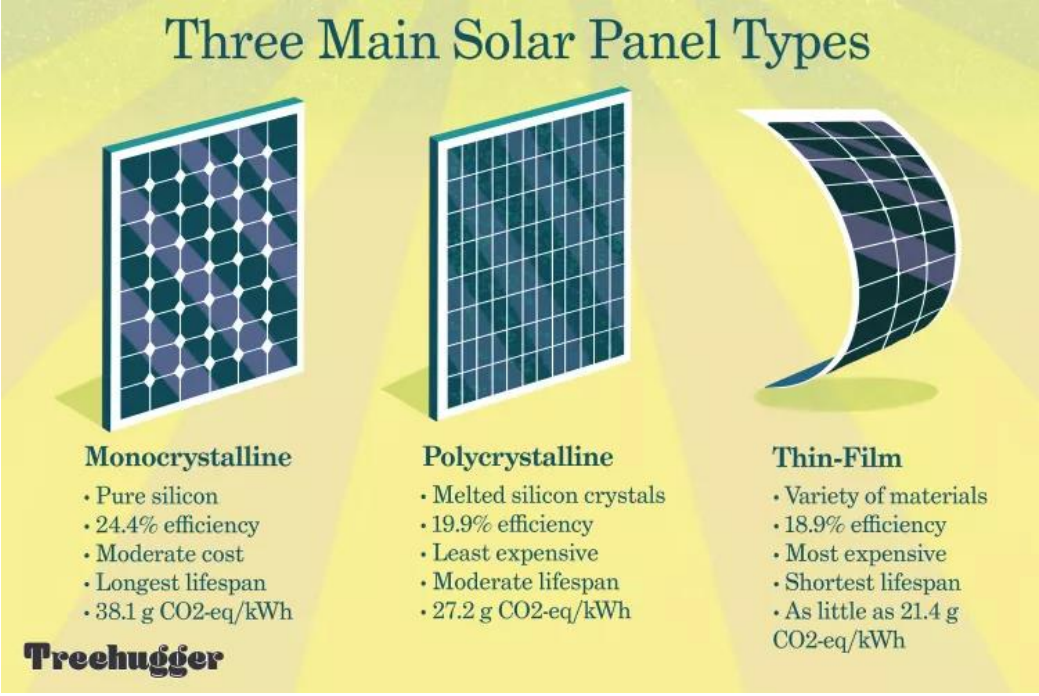


Figure 2 Main Solar Panel Types

The three types shown in figure represent different solar panel module classifications, as categorized by material composition and manufacturing process, cell size, and construction and design.

2.2.1 material composition and manufacturing process

- Crystalline Silicon:
  - Monocrystalline: Made from a single, continuous crystal of silicon.
  - Polycrystalline: Made from multiple silicon crystals fused together.
- Thin-Film: Made by depositing a thin layer of photovoltaic material onto a substrate. Examples include amorphous silicon, cadmium telluride (CdTe), and copper indium gallium selenide (CIGS) [9].

Monocrystalline Silicon Solar Panels:

Single-crystal panels, also known as monocrystalline silicon panels [10], are among the most mature solar energy technologies. Characterized by their consistent dark tint and rounded borders, they are manufactured using high-purity silicon crystals. Monocrystalline panels are renowned for their high efficiency and long lifespan.

Monocrystalline silicon panels are fabricated using individual silicon wafers cut from a single, continuous crystal of high-purity silicon. This process typically involves the Czochralski process, where a seed crystal is slowly pulled from a melt of molten silicon, allowing a single crystal to form. The resulting ingot is then sliced into thin wafers. [3]Each wafer undergoes doping to create the p-n junction, the heart of the solar cell. [4]These cells are then assembled into panels and encapsulated with a protective layer of glass, often coated with an anti-reflective material to maximize light absorption. Due to the high purity of silicon, electrons flow more freely within the cells, resulting in higher energy conversion efficiency compared to other types of solar panels [11].

Polycrystalline Silicon Solar Panels:

Polycrystalline silicon solar panels, also known as multi-crystalline or polysilicon panels, are a widely used type of photovoltaic cell. Polycrystalline panels are manufactured from multiple silicon crystals that have been melted together. They are characterized by a bluish color and a less uniform appearance compared to monocrystalline panels.

Polycrystalline panels are manufactured using silicon wafers produced from multiple silicon crystals. In this process, raw silicon is melted and poured into a square mold, then cooled and sliced into thin wafers. These wafers are assembled into panels, and a protective layer is applied to complete the solar panel. The manufacturing process for polycrystalline panels is generally more cost-effective and simpler due to the use of multiple silicon crystals within each cell.

Thin-Film Solar Cells:

Thin-film solar cells are second-generation photovoltaic (PV) devices fabricated by depositing one or more thin layers of photovoltaic material onto a substrate, such as glass, plastic, or metal. This distinguishes them from traditional silicon-based solar panels, which utilize thick silicon wafers.

Common thin-film materials include Cadmium Telluride (CdTe), Copper Indium Gallium Selenide (CIGS), and Amorphous Silicon (a-Si) [9]. These materials absorb sunlight and convert it into electricity. Due to the thinness of the layers, typically just a few micrometers, thin-film cells can be lightweight and flexible.

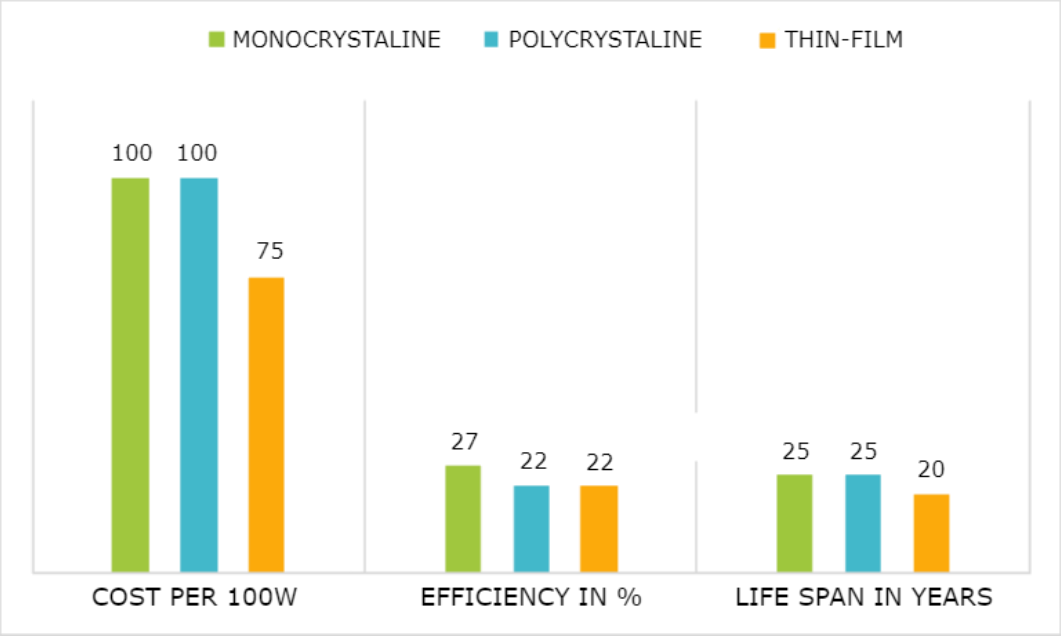


Figure 3: Cost per watt, Efficiency and Life span comparison.

2.3 Electrical proprieties

Understanding the electrical properties of solar panels involves considering key concepts such as V_{oc} (open-circuit voltage), Isc (short-circuit current), MPPT (maximum power point tracking), the role of bypass diodes, and the impact of shading. Analyzing V-I curves is crucial for understanding the electrical behavior of solar cells and modules. It's essential to differentiate between module efficiency and cell efficiency, recognizing that module efficiency is typically lower due to factors like interconnections and shading. While theoretically possible, powering a PV cell with an LED and then using the emitted light to power the same cell would be highly inefficient due to energy losses in both conversion processes. Understanding these concepts is fundamental for effectively interpreting solar panel datasheets, which provide critical information such as V_{oc}, I_{sc}, V_{mp} (voltage at maximum power point), Imp (current at maximum power point), Pmax (maximum power output), efficiency, and temperature coefficients.

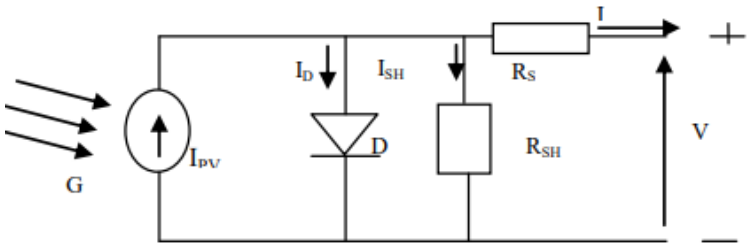


Figure 4: Photovoltaic Model

Where:

- G: represents light of the sun.
- I_{Pv}: the current generated by the photovoltaic
- I_D: the current of the diode.
- R_{SH} : shunting resistor.
- R_S: series resistor.
- V: the output voltage of the photovoltaic.

The two important element that is important in my point of view is R_S and R_{SH} since the other componet manily depends on factor other than the module itself. And the impact of each resistor [13]:

Table 2: mpact of Shunt and Series Resistance in Solar Cells

Resistance Type		R_{SH}	R_S
Concept		Parallel resistance causing current leakage.	Internal resistance hindering current flow.
Impact	Low Ohms	More leakage, lower efficiency.	Minimal drop, higher efficiency.
	High Ohms	Less leakage, higher efficiency.	Voltage drops, lower efficiency.
Causes		- Defects in the semiconductor (cracks, impurities). - Poor contact between metal fingers and the semiconductor.	-Resistance within the semiconductor. - Resistance at the contacts between metal fingers and the semiconductor.

A solar panel module is essentially a series of interconnected photovoltaic cells. [1] Each cell typically produces an average voltage of 0.5 volts. [2] This voltage output gradually degrades over the module's lifespan. Refer to Figure 6 for a better understanding of this concept.

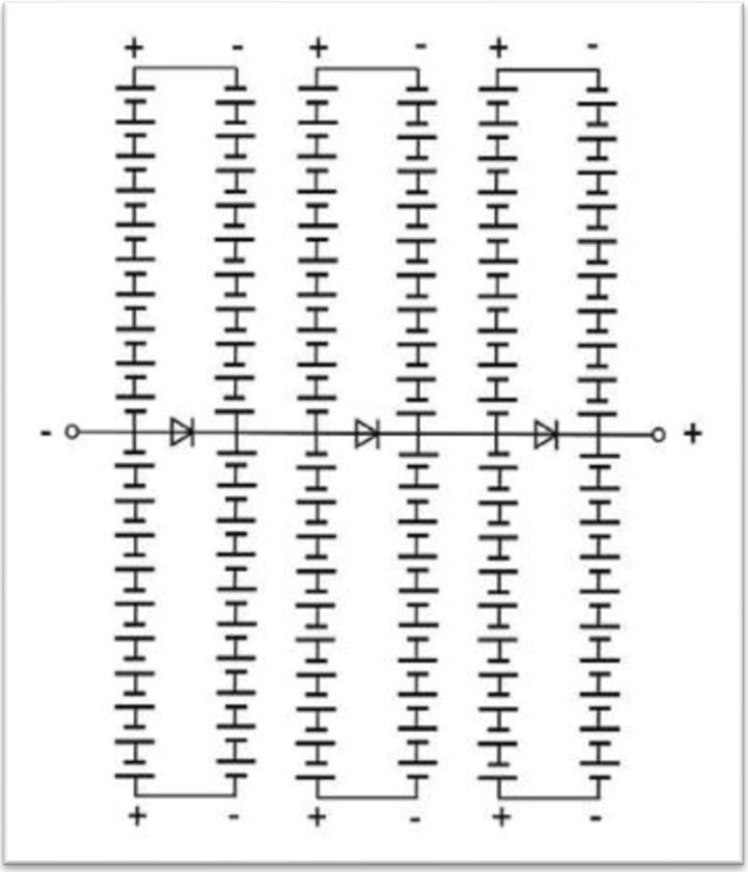


Figure 5: PV, solar panel module.

The last critical concept to grasp before analyzing a datasheet is the role of bypass diodes. Bypass diodes provide an alternative current pathway within a solar panel module, mitigating the impact of shading, accumulated dirt, or dust. When a portion of the module is shaded, the bypass diodes allow current to flow around the affected area, preventing excessive current from flowing through the shaded cells and potentially damaging them.

However, the activation of a bypass diode inevitably leads to power loss. While the diode prevents damage, it also bypasses a portion of the solar cell's output, resulting in a decrease in overall power generation.

While ideally, each cell or even each half-cell within a module would have its own bypass diode for maximum efficiency, this implementation significantly increases manufacturing costs. As a compromise, a typical configuration utilizes three bypass diodes per module. This approach offers a balance between cost-effectiveness and performance, minimizing power losses due to shading while maintaining a reasonable manufacturing cost.

Table 3: Values at Standard Test Condition STC [14].

POWER AT STC	525 W	530 W	535 W	540 W	545 W	550 W
Short Circuit Current - Isc (A)	13.65	13.72	13.79	13.86	13.93	14.00
Maximum Power Current - Impp (A)	12.76	12.83	12.90	12.97	13.04	13.11
Open Circuit Voltage - Voc (V)	49.15	49.30	49.45	49.60	49.75	49.90
Maximum Power Voltage - Vmpp (V)	41.15	41.31	41.47	41.64	41.80	41.96
Module Efficiency - η' (%)	20.3%	20.5%	20.7%	20.9%	21.1%	21.3%

Where STC is (Air Mass AM 15, irradiance 1000 W/m^2, Cell Temperature 25° C).

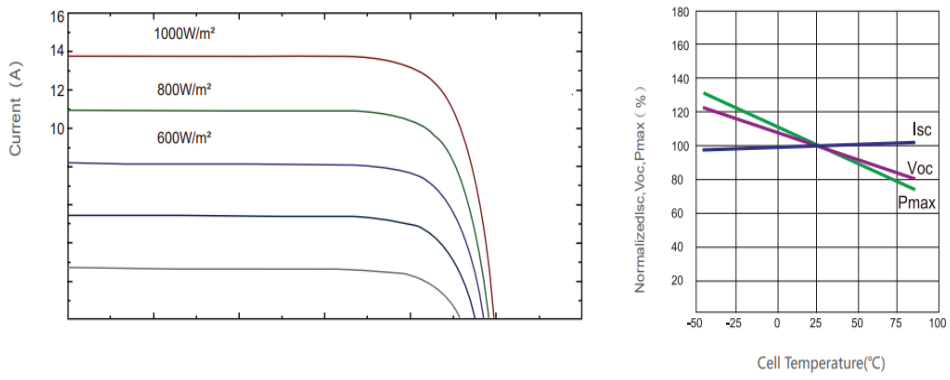


Figure 6: I-V Curves.

2.4 Cleaning Methods for Solar Panels

Conventionally, solar panels are cleaned manually. This approach has several disadvantages including non-uniform cleaning, inaccessibility, potential damage to panels, risk of human casualties (especially in roof-top panels), and difficulties in movement. Additionally, hiring labor for PV panel cleaning can be cumbersome and costly [1]. To overcome these drawbacks, various technologies have been developed worldwide to accomplish cleaning task, and the prevalent cleaning techniques used today are:

1. Robotics: Robotics is an interesting new technology that is being utilized to clean solar panels. One of the most efficient and time-saving approaches is to use robotics to clean solar panels. This results in a more effective cleaning operation that takes less time, lowering the danger of scratches or damage to the solar panel's surface. This will also reduce the amount of damage and injuries that workers may sustain when cleaning panels in hazardous places.
2. Waterless Vibration: It's also simple and doesn't require any special equipment or tools. The fundamental procedure is shaking the surface of your solar panel with a strong equipment known as an industrial vibrator (sometimes referred to as "Vibra-clean") [15]. This approach employs high-frequency vibrations to break down grime and oil on the surface of your solar panel while causing no damage to the surface or interior components. Because of the high frequency vibration, this approach may cause damage to the solar panel.
3. Nanoparticle Coating: Another method for cleaning your solar panels is to use nanoparticle coatings that are specifically intended for cleaning solar panels. These nanoparticle coatings are deposited using an electrostatic spray gun or roller on both metal and glass surfaces. They operate by eliminating dirt and grime from both sides of your solar panels while they are still in their protective containers during storage or shipping, allowing them to be installed quickly after cleaning [16].

These technologies offer several advantages over manual cleaning, including uniform cleaning, accessibility, reduced risk of damage to panels, and reduced risk of human casualties.

2.4.1 Robotics as a cleaning method

Modern innovations have brought about the development of different robotic solar panel cleaners, which include:

Ecoppia E4 which is a fully autonomous robot that uses microfiber brushes to sweep dust off solar panels. This robot is designed for large rows of solar panels installed in dry and sandy environments. Utilizing the effect of gravity, the brushes move downwards while spinning which creates an airflow that helps blow off dust [17]. The robotic cleaner uses an on-board solar panel and battery to store energy which enables the robot to clean at night. Horizontal and vertical translation is made possible using a guide rail and wheels.



Figure 7: Ecoppia E4

Another developed robotic solar panel cleaner is wash panel systems that cleans arrays of solar panels by moving a vertical brush horizontally across rows of solar panels. This system can be deployed automatically. A water hose is attached to the robot for wetting the solar panels while it cleans [18]. Unlike the Ecoppia E4 this system does not require a track.



Figure 8: A Wash panel system

Additionally, there is the hyCleaner Solar facelift cleaning robot that allows for easy, economic, and fast cleaning [19]. This robotic cleaning system is controlled by a radio remote control and draws its power from lithium-ion batteries without an external power source enabling the operator to achieve an optimum cleaning result without physical efforts.



Figure 9: hyCleaner Solar facelift

There are many other advanced solar cleaning systems available that can be used for both small and large solar panel setups to keep them clean and well-maintained. When selecting the best solar cleaning system, it's important to consider factors like the size of the installation, local laws, access to water, and your budget. Regularly, cleaning solar panels with a maintenance system improves their performance and extends their lifespan.

Chapter 3: Methodology And Design Analysis

In this chapter, a comprehensive analysis of the robot's methodology and design is presented. The overall project can be broken into sections such as design and safety considerations, plus the mechanical structure, electrical and electronic circuits, and the software systems. The methodology sections detail the design of the mechanical structure, circuit systems, and the flow chart of the Arduino codes. Furthermore, the hardware used to construct the prototype will be listed down in this chapter.

3.1 Design Considerations

In this project, the cleaning robot is designed to target arrays of solar panels. These solar panels will be arranged into groups or arrays, as Figure 11 shows. The recent common practice of solar panels installation suggested that a pathway should be reserved between the solar panel arrays for future services and maintenance. It is recommended to maintain a clearance around the entire solar panel array and establish a pathway between the rows to ensure safe and easy maintenance [20].



Figure 10: solar panels arrangements.

3.1.1 Stubborn soiling

Solar panels typically suffer from issues such as bird droppings, dust, and watermarks. Therefore, the cleaning robot must be capable of removing stubborn and aggressive forms of soiling, particularly bird droppings.

To address this challenge, the robot will be equipped with a high-speed rotating brush with an appropriate pressure to remove stubborn dirt, especially bird droppings and watermarks. Water will be sprayed onto the solar panels to aid the cleaning process.

3.1.2 Gaps between the arrays and the inclined surface

The targeted solar panel site employs an array arrangement, which includes gaps between the arrays. These gaps present a challenge for the robot to move across to the next array. The solution lies in selecting the appropriate dimensions for the powertrain method, which is tracked. Additionally, the solar panels are installed at specific tilting angles, making the material of the tracked belt crucial.



Figure 11: Gaps between the arrays.

3.1.3 The weight of robot on the solar panel surface

When designing a cleaning robot for solar panels, the weight of the robot on the solar panel surface is a critical consideration. Solar panels are designed to withstand specific loads, and excessive weight can cause damage or reduce efficiency. Key considerations include:

1. Load Capacity: Ensure the robot's weight is within the solar panels' load limits to prevent damage.

2. Material Selection: Use lightweight materials like aluminum, carbon fiber, or certain plastics to keep the robot's weight low.

3. Weight Distribution: Distribute the robot's weight evenly using a well-designed chassis and multiple contact points or tracks

3.1.4 Mobility, Navigation, and Energy efficiency

To effectively clean solar panels, the robot must navigate various inclines, slopes, and obstacles such as frame edges or cables, while ensuring energy-efficient operation. Key considerations include:

1. Optimal Path Planning: by Reducing unnecessary movements to conserve power and ensure efficient cleaning.

2. Obstacle Navigation: Equip the robot to handle physical challenges like panel inclines and obstacles without compromising its cleaning ability.

3.1.5 Safety Considerations

Proper design and safety mechanisms are essential to prevent issues like short circuits, overheating, and explosions, especially in robots that operate in sensitive or hazardous environments [21].

3.2 Mechanical Structure and Design

This section will provide a detailed analysis of the mechanical structure of the robot. It will cover various design options and discuss how these choices influence the robot's capabilities and performance.

3.2.1 Solar Panel Stress Analysis

To determine the maximum weight of the robot that the solar panels can handle, by ensuring that the stress exerted by the robot does not exceed the yield stress of the glass.

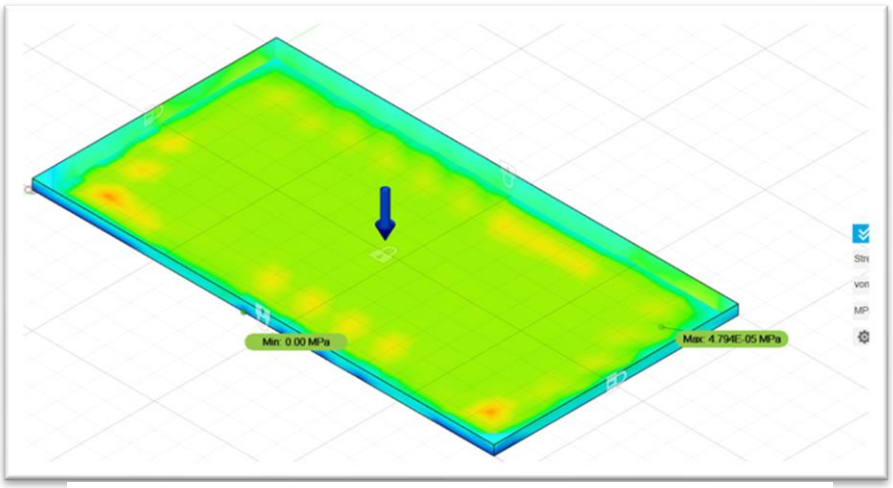


Figure 12: Von Mises stress on glass, by 70N.

After applying 70N as an approximate weight for the robot. The von mises stress on glass is equal to 47.2 Pa, the yield stress of the glass is equal to 33MPa, so it indicates that mobile robot will not fracture the glass.

3.2.1.1 Maximum allowable weight

To find the maximum force (which corresponds to the weight of the robot):

$$Force = Stress \times Area \tag{3.1}$$

Assuming the area of contact between the robot and the glass, as viewed from the top, is approximately 0.01 m² (based on the area of the timing belt).

the maximum force F_{max} the glass can handle is:

$F_{max} = 33{,}000{,}000\text{Pa} \times 0.01\text{m}^2 = 330{,}000\text{N}$

The Maximum weight for the robot is equal to the maximum force F_{max} the glass can handle. The maximum weight of the robot that the solar panels can handle is **330 kN**.

This calculation ensures that the stress exerted by the robot does not exceed the yield stress of the glass, keeping the panels safe from fracture.

3.2.1.2 The normal tensile and compression stresses.

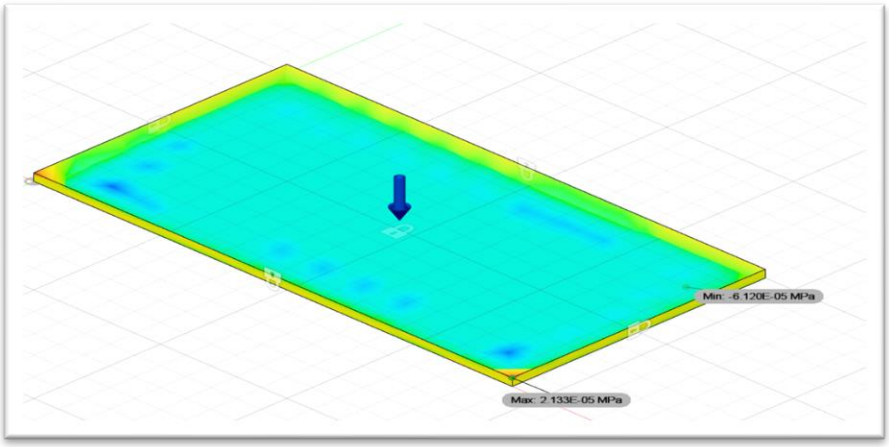


Figure 13: Normal stress on solar panel.

3.2.1.3 The strain.

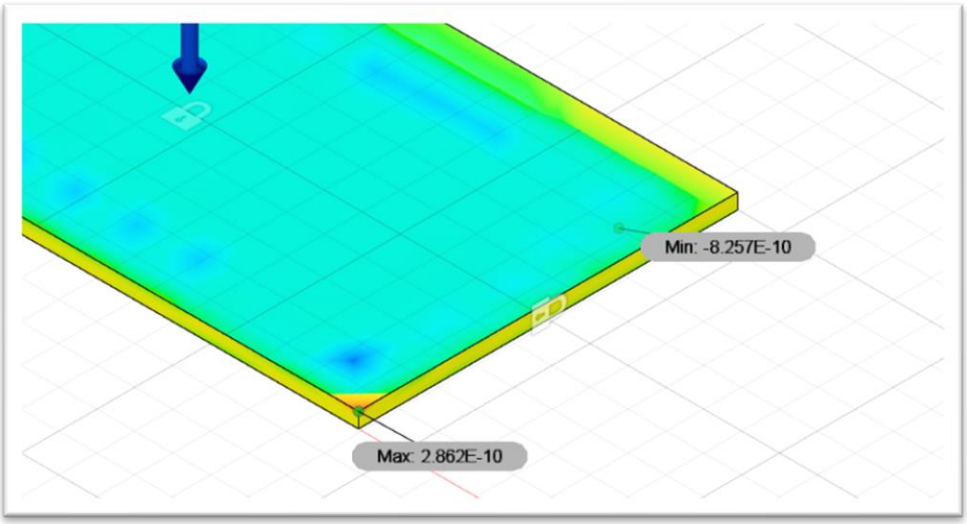


Figure 14: The strain on the solar panel.

3.2.2 Chassis Design and Material Selection

The chassis is the core structure of a mobile robot, providing essential support and stability. This part will explore the design of the chassis, focusing on choosing the right materials and performing stress analysis to ensure it is both strong and lightweight.

3.2.2.1 Material Selection

Choosing the right material for the right job is paramount as it impacts the robot's durability, weight, and functionality. Common materials include:

- Aluminum: Lightweight and durable, ideal for most applications.
- Stainless Steel: Used in industrial settings where strength is a priority.
- Plastic: Suitable for lightweight and indoor robots.

Material selection is a complex task that requires understanding material properties, environmental conditions, costs, performance, and the design process [22].

Design Requirements and Selection Criteria:

Mechanical properties involve ensuring the material has the necessary strength, stiffness, and durability to handle operational stresses. Weight is important because lightweight materials improve mobility and energy efficiency. Environmental resistance means choosing materials that can resist moisture, temperature changes, and corrosive substances. Manufacturing constraints include considering the availability, cost-effectiveness, and compatibility of materials with your design requirements.

Identify and Evaluate Candidate Materials:

Table 4: Candidate Materials

Material	Advantage	Disadvantage
Aluminum	Lightweight, strong, easy to shape, resistant to corrosion	Can be more expensive than other materials
Steel	Durable, rigid, can withstand high stress or impact	Heavy, can be challenging to work with without proper tools
Plastics	Lightweight, flexible, easy to fabricate	May not be as durable or strong as metals
Wood	Cheap, abundant, easy to work with	Not as durable or weather resistant as metals or certain plastics
Fiber composite	Lightweight, high strength-to-weight ratio	Can be expensive and more difficult to work with than other materials

The mechanical properties of the materials:

Aluminum:

- Ultimate Tensile Strength: 90 MPa (pure), 600 MPa (alloys)
- Yield Strength: 11 MPa (pure), 400 MPa (alloys)
- Modulus of Elasticity (Young's modulus): 70 GPa (pure), 70-79 GPa (alloys)
- Hardness: Brinell hardness is approximately 240 MPa

Steel:

- Ultimate Tensile Strength: Varies greatly depending on the specific type of steel, but it's generally very high.
- Yield Strength: Also varies, but it's typically high, making steel very resistant to deformation.
- Modulus of Elasticity (Young's modulus): Around 200 GPa, which means steel is very stiff.
- Hardness: Varies, but steel is generally very hard and resistant to scratching.

ABS Plastic:

- Known for its high impact resistance and toughness, even at low temperatures.
- Good mechanical properties, including high strength and stiffness.
- Exhibits good electrical insulating properties.

Wood:

- Mechanical properties can vary greatly depending on the species and the part of the tree from which the wood is taken.
- Generally, has good tensile and compressive strength parallel to the grain.
- Relatively weak perpendicular to the grain and can be quite brittle depending on the species and treatment.

Fiber Composite (like Kevlar or Carbon Fiber):

- High strength-to-weight and stiffness-to-weight ratios.
- Generally resistant to chemicals, UV radiation, and heat.
- Mechanical properties depend on the fiber, matrix, interfacial strength, fiber dispersion, and fiber orientation.

Based on the mentioned steps, design requirements, and material selection criteria, aluminum has been chosen for the robot's chassis construction because it met the following requirements:

- Appropriate mechanical properties
- Light weight
- Resistant to environmental conditions

3.2.2.2 Material study

By analyzing the resulting stress and strain, it can be determined if the design will withstand the applied forces without failure, according to the stress-strain curves.

First, a force of 70N is applied to the body, which is derived from its weight.

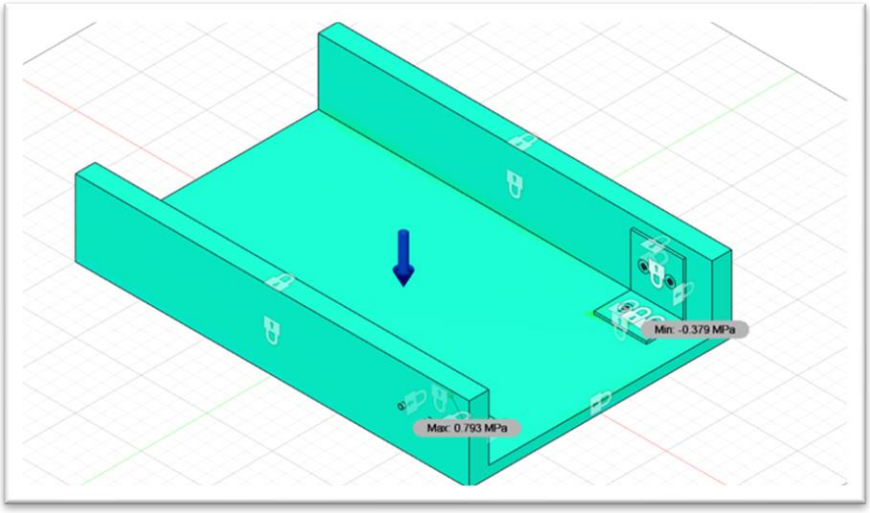


Figure 15: A force of 70N applied to the cart.

The analysis indicates that the highest stress will occur at one of the screws.

- Highest tension stress will equal 0.793MPa.
- Highest compression stress will equal -0.379MPa.

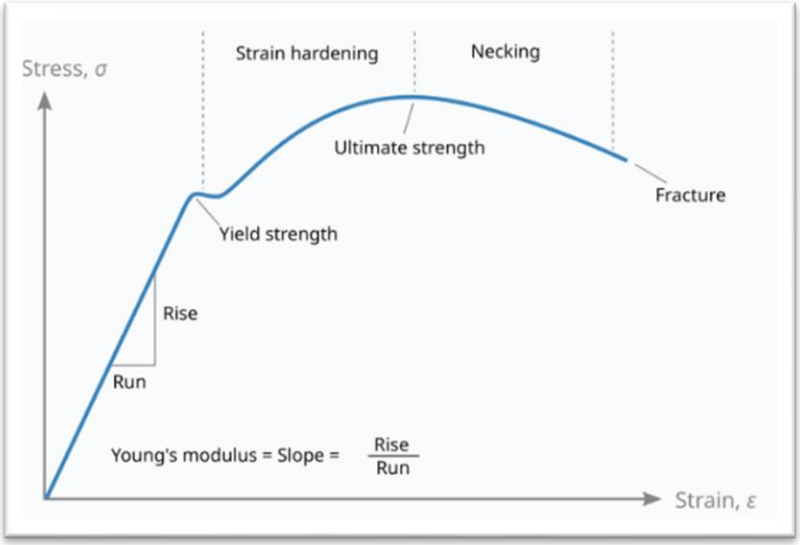


Figure 16: stress strain diagram.

Also, there are more theories about stress analysis, one of them is von-mises.

Von Mises stress is a scalar value used in engineering and material science to predict yielding of materials under complex loading conditions. It is particularly useful because it reduces a multi-dimensional stress state into a single equivalent stress, allowing engineers to compare it to a material's yield strength.

- If σ_v is less than the yield point, the material will remain in the elastic region.
- If σ_v exceeds the yield point, the material will undergo plastic deformation.

Von Mises stress is always positive and equals to 0.556Mpa.

The yield stress for Aluminum equals 275MPa, which means no failure will occur.

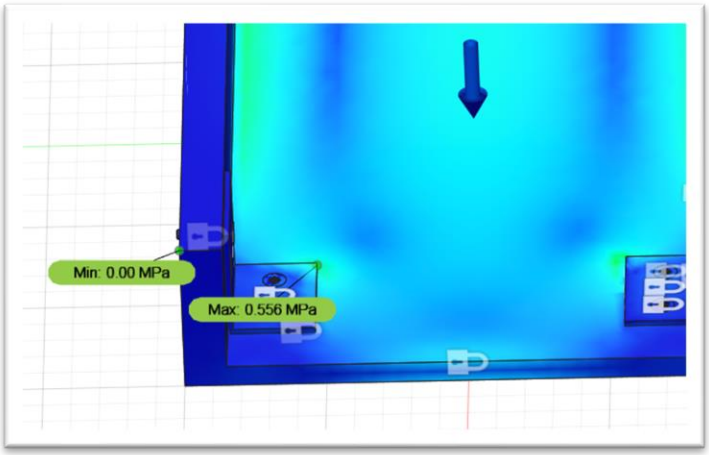


Figure 17: Von Mises stress.

Now let’s look at the strain:

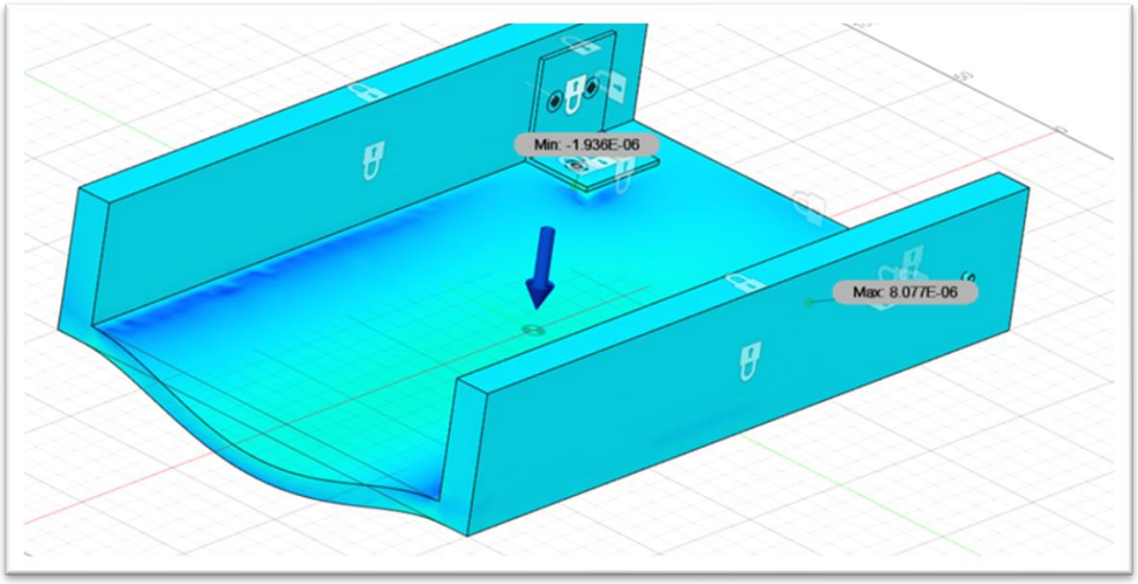


Figure 18: The strain on the chasse.

The strain will take on this shape; however, due to its small magnitude, a scale was used to visualize the deformation. This scaling allows for a clearer understanding of how the deformation manifests, despite the minimal strain.

One of the factors that should be considered too is the thickness of the material. Thicker specimens can distribute stress over a larger volume, which might reduce the fracture from surface defects.

- Minimum strain will be equal to -1.936 micro, caused by compression.
- Maximum strain will equal to 8.077 micro, caused by tension.

3.2.3 Screws

The screws will get effected by normal stress, bearing stress and shear stress. Since the robot’s weight is approximately 70N and there are 12 screws across all the mobile robots. The force per screw will equal 5.833N.

The selected screw type is M4 fine pitch screws. Their detailed specifications are in table 6.

Table 5: M4 fine pitch screws specifications

Nominal Size	d=D	p	r	d2=D2	d3	D1	h3	H1
M4x0.5	4	0.5	0.072	3.675	3.387	3.459	0.307	0.271

All dimensions stated in are mm (millimetre)

The axial stress in the body of the screw due to load *Fn:*

$$\sigma = \frac{F}{\frac{\pi}{4}(d - 0.9382p)^2} \tag{3.2}$$

Normal stress will equal to 0.6MPa.

The shear stress occurs when a force acts parallel to the cross-sectional area of the screw. The parallel force caused by friction can be found by this:

$$F_t = \mu F_n \tag{3.3}$$

μ is the static friction coeffeicnet between aluminum and steel = 0.61.

$$F_t = 3.55813 \text{ N}$$

Shear stress will equal to = 0.28Mpa.

The bearing stress also depends on Ft and the thickness of the plate.

$$\sigma_b = \frac{F_t}{t \cdot d} \tag{3.4}$$

Where t is the thickness, and d is the diameter. $\sigma_b = 0.44Mpa$.

The Poisson's ratio for steel 4.6 is equal to 0.3.

From this ratio, the lateral strain can be found using this formula:

$$v = -\frac{d\varepsilon_{trans}}{d\varepsilon_{axial}} \tag{3.5}$$

The lateral strain will be -0.855×10^{-6} .

3.2.4 Powertrain Selection of the Mobile Robot:

In the process of constructing a mobile robot, the selection of the powertrain is a pivotal decision. The powertrain dictates the robot's movement and navigation capabilities, thereby influencing its overall performance. Let's delve into the various types of powertrains you might consider:

- Differential: This type is widely used due to its simplicity and ease of control.
- Omnidirectional: This type allows movement in all directions.
- Tracked: This type is ideal for uneven terrains.

In certain scenario, such as dealing with friction and stress, a tracked powertrain can offer several advantages for a robot [23]. Here's why:

1. **Enhanced Traction**: Tracked powertrains have a larger contact area with the ground compared to wheels, which increases traction. This can be particularly beneficial on surfaces like powdery or icy soil.
2. **Uniform Weight Distribution**: Tracked robots are typically lighter than traditional robots, which allows their weight to be distributed over a larger surface area, thereby exerting less pressure on the ground. This can prevent the robot from sinking into soft surfaces like sand or getting stuck in mud.
3. **Superior Obstacle Negotiation**: Tracked robots can effectively handle obstacles and uneven surfaces. The tracks can conform to the shape of the obstacle, enabling the robot to climb over it.
4. **Reduced Stress**: The enhanced traction and ability to negotiate obstacles can result in lower mechanical stress on the robot's components.
5. **Stability**: Tracked robots are generally more stable than wheeled or legged robots, reducing the risk of tipping over.

While tracked powertrains can be more complex and require more maintenance than other types, these benefits make them an excellent choice for a wide range of applications, particularly in challenging environments. However, the choice of powertrain should always be based on the specific requirements of the robot and its intended use.

3.2.4.1 Gear wheel and Timing belt

To simplify the process of selecting the gear wheels and timing belt, a practical diameter was assigned to the gear wheels, which were to be lathed. Selecting a timing belt drive requires compiling all relevant design parameters, including:

a. Type of driver, shaft speed, and power to be transmitted.
b. Type of driven machinery, shaft speed or drive ratio.
c. Approximate shaft center distance required.
d. Number of hours daily, drive will be used, and any special operating conditions (temperature, abrasive dust, etc.).
e. Shaft diameters and any space restrictions affecting pulley diameters or width.

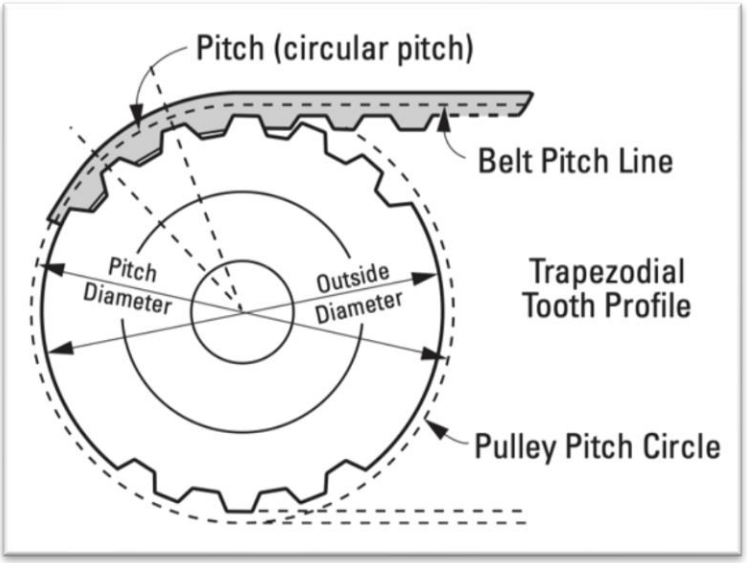


Figure 19: Belt and Wheel Mechanism.

3.2.4.2 Timing belt:

Calculate the Design Power

The design Power Pd is determined by multiplying the transmitted motor power P by the application factors f1, f2, f3 and f4 as applicable.

$$P_d = P.(f_1 + f_3 - f_4).f_2 \tag{3.6}$$

Where Application Factor (f_1) (Service Factor) relates the type of driver and driven equipment to daily usage. Application Factor (f_2) (Speed Increasing Drives) is relevant

to the respective speed increase ratio. For Application Factor (f_3), if an idler is used, add 0.2. For Application Factor (f_4), if machinery is only used intermittently or seasonally, deduct 0.2 from the service factor.

Determine Belt Length and Center Distance by using the equation below:

$$L = 2C + 2\pi r + \frac{(r_1 - r_2)^2}{C} \tag{3.7}$$

Since the diameter of both pulleys are 10 cm this equation can be reduced to:

$$L = 2C + 2\pi r \tag{3.8}$$

Where C is the distance between the center of the first wheel and the center of the second wheel. And it is approximately equals to 22.5 cm

So, the required length of the belts equals $L = 73.4$ cm.

The design of pulleys depends on the power of the motor that will drive it. The motor has a rated 60 W but in the design process, some factors should be considered (f1, f2, f3, f4).

$$P_{design} = P_{rated} . (f_1 + f_3 - f_4) . f_2 \tag{3.9}$$

Table 6: service factor -f1.

DRIVEN MACHINE (LOAD)		TYPES OF DRIVER					
		'SOFT STARTS'			'HEAVY STARTS'		
The machines listed below are representative examples only. Select the class whose load characteristics most closely approximate those of the machine being used. For drives with heavy pulsating or high shock loads, consult Cross & Morse.		Electric Motor AC - Star Delta start DC - Shunt Wound Internal Combustion Engines and other prime moves fitted with Centrifugal Clutches. Dry or Fluid Couplings or Elecronic Soft Start Start Devices.			Electric Motor AC - Direct-on-Line start DC - Series and Compound Wound Hydraulic Motors Internal Combustion Engines with Mechanicl Drive Prime movers not fitted with soft start devices		
Load Type	Typical Machinery	Operating Hours per Day			Operating Hours per Day		
		under 10	10-16	Over 16	under 10	10-16	Over 16
Smooth	Business Equipment. Light Domestic Machinery, Tachometers, Camera and Radar Equipment	1.0	1.2	1.4	1.2	1.6	1.8
Light Shock	Liquid Agitators, Centrifugal Pumps and Compressors, Uniformly Loaded Belt Conveyors, Fans up to 7½ kW, Calenders. Rotary Screens, Dryers, Exhausters, Woodworking Machinery.	1.3	1.4	1.6	1.6	1.8	2.0
Medium Shock	Mixers, Belt Conveyors. Fans over 7½ kW, Generators, Vibrating Screens, Augers, Granulators, Laundry and Printing Machinery, Machine Tools.	1.5	1.7	1.9	1.9	2.0	2.2
High Shock	Bucket, Pan and Screw Conveyors, Reciprocating Pumps and Compressors, Exciters, Paper and Textile Machinery.	1.7	1.9	2.0	2.0	2.2	2.3
Heavy Shock	Mills, Crushers, Hoists, Dredge Pumps, Brickwork and Rubber Machinery, Oil Field Equipment.	1.9	2.0	2.1	2.2	2.3	2.4

This table shows different types of loads and its factor of safety, in this project f1 can be considered to be equal to 2.

Table 7: Speed Increase Factor -f2.

Speed Increase Drive Ratio	1.00 to 1.24	1.25 to 1.74	1.75 to 2.49	2.50 to 3.49	3.50 and above
Factor f2	1.00	1.06	1.12	1.18	1.25

since the speed ratio is one, it shows that f2 is equal to 1.

f3 depends on using idler, if idler is added, f3 will be 2 otherwise it will be zero.

f4 depends on the machinery, if machinery is only used intermittently, f4 will be equal to 2.

$$P_{design} = P_{rated}.(f_1 + f_3 - f_4).f_2 = \ 0.12\text{KW}$$

With a rated motor speed of 90 RPM, the selection of the series can be done as follows:

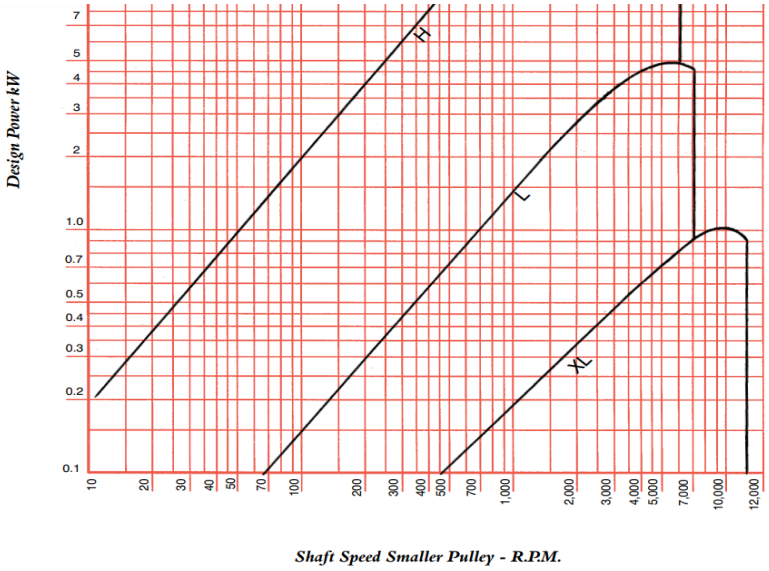


Figure 20: Design Power – Shaft Speed Curve.

Based on the design power and the rated speed of 90 RPM, the L series would be suitable. However, to ensure better safety alongside market availability, the XL series will be used.

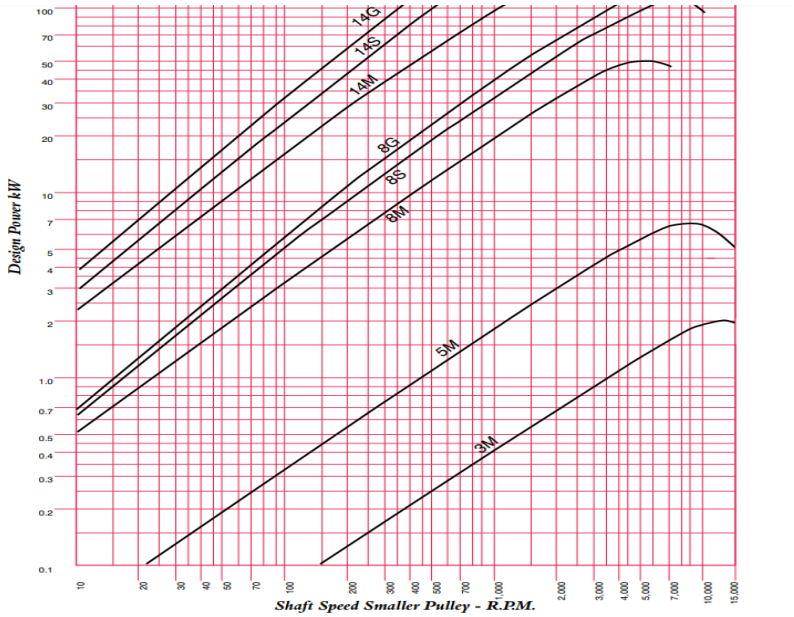


Figure 21: Design Power – Shaft Speed Curve. (metric series)

The same concept applies to the metric series. The specifications lead to the 5M series, but selecting the 3M series at constant power will result in a safer choice.

3.2.4.3 Gear wheels:

Designing the gear wheels, which is to be lathed: since the selected gear wheel has a diameter of 10 cm and the type of the belt is XL (pitch = 0.2 inch or 0.508 cm), the number of teeth on the pulley can be determined by this equation:

$$N = \frac{\pi.D}{P} \tag{3.10}$$

Where (N) is the number of teeth, (D) is the diameter of the wheel (10 cm), and (P) is the pitch of the belt (0.508 cm). By substituting the values, the number of teeth is equal to 60.

A pulley in the XL series with 60 teeth is needed. Both the belt and the gear wheels have a pitch of 0.508 cm, ensuring compatibility. Therefore, a 60XL037 can be purchased or manufactured with a different type of metal to the same specifications as the 60XL037.

The gear wheels are manufactured with Polytetrafluoroethylene (Teflon) due to several reasons, such as less friction, light weight, ease of movement on smooth surfaces, and resistance to chemicals.

3.2.5 Actuators used in this project

This mobile robot features a unique actuator setup specifically designed to meet the project's requirements. This setup is part of a 2-degree-of-freedom (DOF) manipulator, which is a critical component of our robot's design.

- 2-Degree-of-Freedom (DOF) Manipulator

The robot's manipulator has two independent movements or axes of rotation. This allows the manipulator to reach different positions and orientations in a two-dimensional space, providing the flexibility needed for our robot's tasks.

- Motor-Driven Circular Brush

The manipulator is equipped with a motor-driven circular brush. The brush rotates, driven by a motor, to sweep or scrub surfaces. In our case, it's used for cleaning solar panels, ensuring they operate at maximum efficiency.

The combination of a 2-DOF manipulator, a motor-driven circular brush. A yaw joint allows the robot to effectively and efficiently clean solar panels, even in challenging conditions.

After doing some stability calculations, Further details about the manipulator motors can be found in the following section.

3.2.5.1 Robot's stability

To demonstrate that the 2-Degree-of-Freedom (DOF) Manipulator will not cause the mobile robot to flip, the stability of the system can be analyzed using static equilibrium principles.

Identification of Forces and Torques:

- Robot weight (W_{robot}): Acts vertically downward through the center of gravity (CoG) of the mobile robot.
- Arm weight (W_{arm}): Acts downward at the arm's CoG.

To Calculate the Moment about the Tipping Axis:

$$\tau_{arm} = W_{arm} \cdot d_{arm} \tag{3.11}$$

$$\tau_{robot} = W_{robot} \cdot d_{robot} \tag{3.12}$$

d = d_{arm} = d_{robot} = 58mm

W_{robot} = 9.81*6.4 Kg = 62.784 N

W_{arm}= 9.81*.6 Kg = 5.886 N

Check Stability Condition:

For the robot to remain stable, the stabilizing torque must be greater than or equal to the destabilizing torques: $\tau_{robot} \geq \tau_{arm}$.

3.641472 ≥ 0.341388

So, the system will remain stable, since the robotic arm is light compared to the mobile robot.

3.3 Electrical and Electronic Circuit Design

Following the mechanical structure design, the second section of the project involved designing the electrical and electronic circuits. These circuits are designed to control multiple DC motors and servos using an ESP32 microcontroller and L298N motor drivers. The system is powered by two sets of three 18650 Li-ion batteries and an additional pair of 18650 Li-ion batteries.

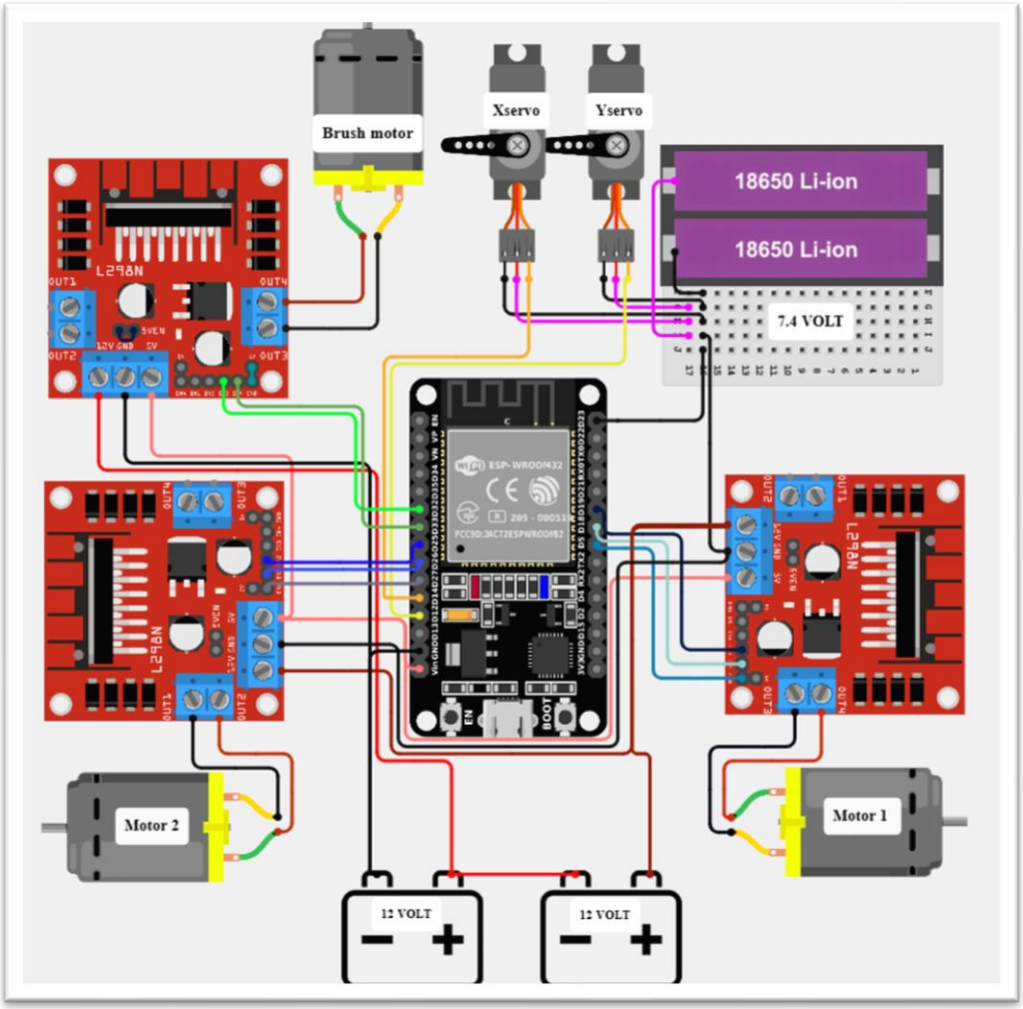


Figure 22: Wiring diagram.

This circuit offers a robust and versatile solution for controlling multiple DC motors and servos with an ESP32 microcontroller. The thoughtful integration of components ensures reliable operation and ease of control.

3.3.1 Centralized Control

The controller is the computational unit of the robot, responsible for processing sensor data and making decisions. It's essentially the brain of the operation and can range from simple microcontrollers to complex embedded systems. The microcontroller is a small, self-contained computer that is used to control the robot system [24]. It typically has a CPU, memory, and input/output (I/O) peripherals and is programmed to perform specific tasks. The microcontroller is the brain of the robotic solar panel cleaner, responsible for controlling all of the other components and making decisions based on received input.

In this project, the ESP32 was chosen for its robust capabilities and versatility [25]. Here are the reasons for selecting the ESP32:

- Processing Power: Dual-core (or single-core) 32-bit Xtensa LX6 microprocessor, capable of operating at either 160 or 240 MHz
- Wireless Connectivity: Bluetooth v4.2 BR/EDR and BLE, enabling control of the robot via the microcontroller.
- Cost: Approximately $5 to $10 per unit, making it a cost-effective choice
- Power Management: Ultra-low-power co-processor, ideal for battery-powered applications

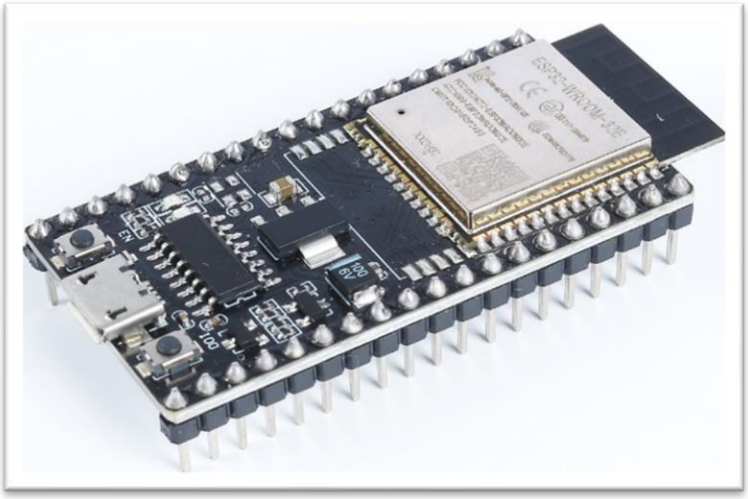


Figure 23: ESP32 module.

- Authentications: IEEE 802.11 standard, incorporating features like WPA, WPA2, and WPA3.

- Peripheral Interfaces: 34 GPIO pins, supports Pulse Width Modulation PWM for motors and servos, with up to 16 channels available. Communication interfaces are also included.

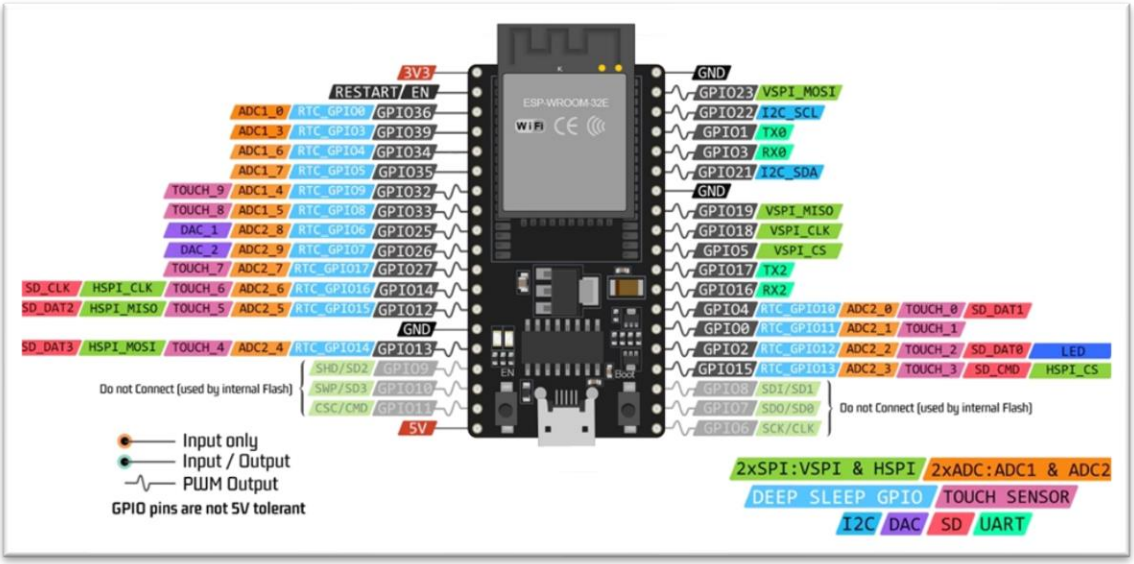


Figure 24: ESP32 Pinout.

The ESP32 was used in this project to handle control logic, sending signals to the motor drivers and servos to achieve the desired motion and functionality.

Arduino IDE was used to write and compile the ESP32 codes. It was also used to upload the written codes to the ESP32. Furthermore, the Arduino IDE provided a serial monitor function, which helped trace the condition of the program during practical testing and debugging tasks. Further details about the software can be found in section 3.4.

3.3.2 Motor Drivers

The motor driver manages the operation of the motors used in the movement of the robot. It acts as an interface between the robot's control system and the motors to enable control and coordination of speed, direction and torque for the robotic system.

In this project, two L298N motor drivers are used to control the DC motors. Additionally, a third L298N motor driver serves dual purposes:

- **Relay**: It acts as a relay for the brush motor.
- **Voltage Regulator**: It functions as a voltage regulator, providing a 5-volt power supply derived from a regulated 12-volt source.

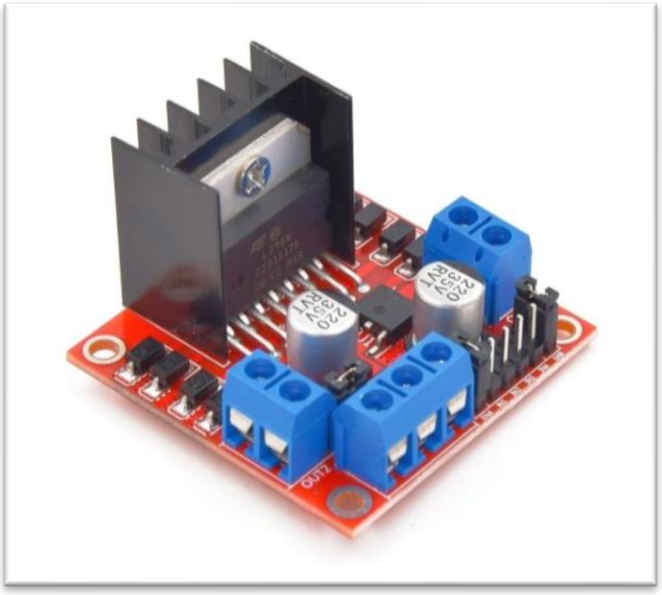


Figure 25: L298N motor driver.

There are several advantages and features for using the L298N DC Motor Driver [26] in PanelGaurd Robotic System:

- **Built-in Protection**: Features like overcurrent protection and thermal shutdown help safeguard the driver and connected components from damage.
- **Wide Voltage Range**: It supports a voltage range from 5V to 35V, making it compatible with various power sources

- **Dual H-Bridge Design**: This allows for independent control of two motors, enabling forward and reverse operation as well as speed control through Pulse Width Modulation (PWM).

- **High Current Capacity**: The L298N can handle up to 2A per channel, which is sufficient for many motors control applications

In this project, two separate L298N motor drivers were used to control the two DC motors that run the wheels of the robot, with each motor connected to its own driver. This setup was necessary because each motor has a high current draw, close to the rated current capacity of the L298N, which is 2A per channel. By using one driver per motor, reliable operation is ensured without overloading the drivers. Additionally, this approach helps manage heat dissipation more effectively, as each driver can generate significant heat when operating near its maximum capacity. To ensure safety, each driver is equipped with a heat sink and features built-in thermal shutdown to prevent overheating.

Furthermore, a third L298N motor driver serves dual purposes: it acts as a relay for the third motor and functions as a voltage regulator, providing a 5-volt power supply derived from a regulated 12-volt source. This configuration optimizes power distribution and control across the entire system, ensuring both efficiency and safety.

3.3.3 Wheels motors

The robot should be able to move at a speed of 0.2 m/s, have wheels that are 10cm in diameter, have a maximum carrying capacity of 7 kg, and be outfitted with two drive wheels

- Motor sizing

The motor was selected and sized based on detailed calculations, as outlined below. This is the Free Body Diagram (FBD) of the robot on the panel.

Assumptions:

- There is no slip on the primary timing pulleys and the timing belt.
- The tensions of the timing belts on both sides of the cleaning robot are the same.
- The slip resistance is not affected by the direction of rotation of the broom

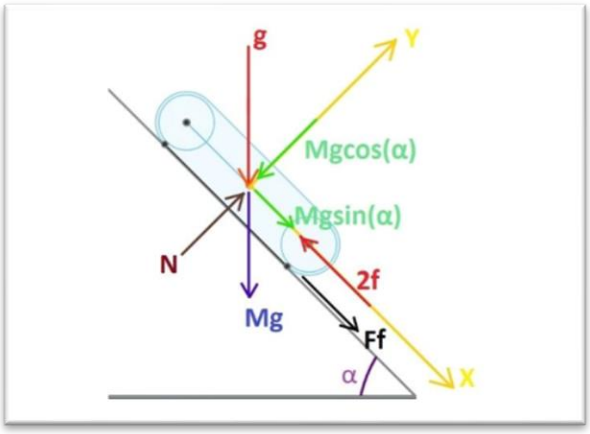


Figure 26: FBD of the robot on PV.

Applying Newton's second law on the X-axis and Y-axis. Considering the cleaning robot in the stationary state on the PV panel with a tilted angle, worst-case scenario for operation is on an inclined glass surface at a 45-degree angle.

$$\sum F_x = 0 \tag{3.13}$$

$$2f - (M)gsin(45) - F_f = 0$$

Where $2f$ are the input forces from two motors, and M is the total mass of the robot, and F_f is the frictional force.

To calculate the frictional force the following formula is used:

$$F_f = N\mu_s \tag{3.14}$$

where μ_s is the coefficient of friction, which is 2+ according to table 8, and N is the reaction force. From the summation of forces on y-axis $\sum F_y = 0$.

$$N = (\mathrm{M})g\cos(45) \tag{3.15}$$

Table 8: coefficients of frictions

Material	Coefficient of Static Friction μ_S	Coefficient of Kinetic Friction μ_S
Rubber on Glass	2.0+	2.0
Rubber on Concrete	1.0	0.8
Steel on Steel	0.74	0.57
Wood on Wood	0.25 – 0.5	0.2
Metal on Metal	0.15	0.06
Ice on Ice	0.1	0.03
Synovial Joints in Humans	0.01	0.003

According to the no-slip condition, the greater the static friction, the larger the slip resistance. On the other hand, the static friction force must be larger than the cleaning robot's gravity to ensure the no-slip condition of the cleaning robot.

Additionally, the mass of the robot is 7 kg, and the acceleration due to gravity is approximately 9.8 N/kg. Therefore, with the equation of the frictional force (F_f), substituting it into the equation of $\sum F_x = 0$ and rearranging for the input force from the motors (f):

$$f = \frac{\sqrt{2}}{4} Mg(1 + \mu_s) \tag{3.16}$$

Hence, the total force required from single motor for the robot to move:

$$f = \frac{\sqrt{2}}{4}(7)(9.8)(1 + 2.15) = 76.5 \text{ N}$$

- Torque Requirements

To select a motor that can provide the required force. The torque τ provided by the motor can be calculated using the formula $\tau = f \cdot r$ where f is the force required by the mobile robot, and r is the radius of the wheel. using wheels with a radius of $r = 0.05\ m$. So, the motor torque with factor of safety of 1.5 should be provided is:

$$\tau = f \cdot r \ = (76.5\ N)(0.05\ m)(1.5) = 5.73\ \ N.m \qquad (3.17)$$

Therefore, the motor's rated torque must be 5.73 N.m or greater to meet the required torque with the safety factor applied.

- Motor selected

In the PanelGuard robot, two brushed DC motors were utilized to power the wheels. The 'TAKANAWA 555 Metal Gear' Motor was selected after a thorough evaluation of various considerations, including motor sizing, cost, robustness, durability.

The TAKANAWA 555 Metal Gear Motor is a robust and reliable motor designed for a wide range of applications. Operating at 12V-24V with a speed of 80RPM, this motor offers high torque and efficient performance, making it ideal for tasks requiring precise and powerful motion control.

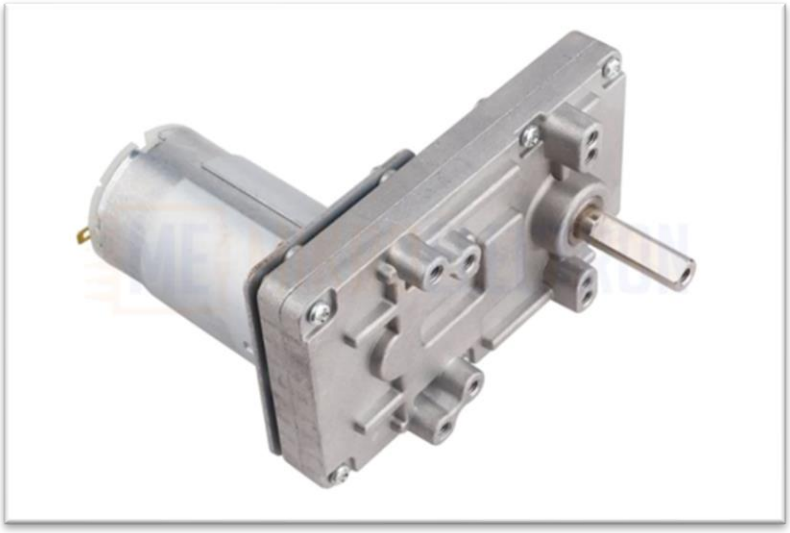


Figure 27: 'TAKANAWA 555 Metal Gear' Motor.

The motor offers an excellent balance between cost and specifications, making it an ideal choice for the project. It features a sturdy metal gearbox that enhances durability and extends the motor's lifespan. Designed to withstand harsh environments and heavy-duty usage, the motor ensures long-lasting performance.

- To estimate the rated power of TAKANAWA 555 motor:

$$P = I \cdot V \tag{3.18}$$

Power (W)=Voltage (V)×Current (A), the motor is supplied with a 24V DC voltage and has a rated current of 2.5A.

$$P = 24\text{V} \times 2.5\text{A} = 60\text{W}$$

- To estimate the rated torque of TAKANAWA 555 motor:

$$P = \tau \cdot \omega \tag{3.19}$$

Power (W) = Torque (Nm) × Angular Velocity (rad/s); the motor has a rated speed of 80 rpm.

$$\tau = \frac{60\text{W}}{80 \times \frac{\pi}{30}} = 7.17\ \text{N.m}$$

As a result, the motor's rated torque is calculated to be 7.17 N.m, which is greater than the required torque of 5.73 N.m, including the applied safety factor.

Choosing a motor with a higher torque than needed means it will work more reliably, last longer, and can be used in more projects. It also provides a safety buffer for unexpected situations. While it doesn't cost more, the benefits often make it worth it.

3.3.4 Arm motors

Two servo motors were implemented to enable the arm to perform two distinct movements. The first servo, ServoY, is responsible for the vertical movement of the entire arm (up/down). The second servo, ServoX, is responsible for the horizontal movement of the second link and the brush (right/left).

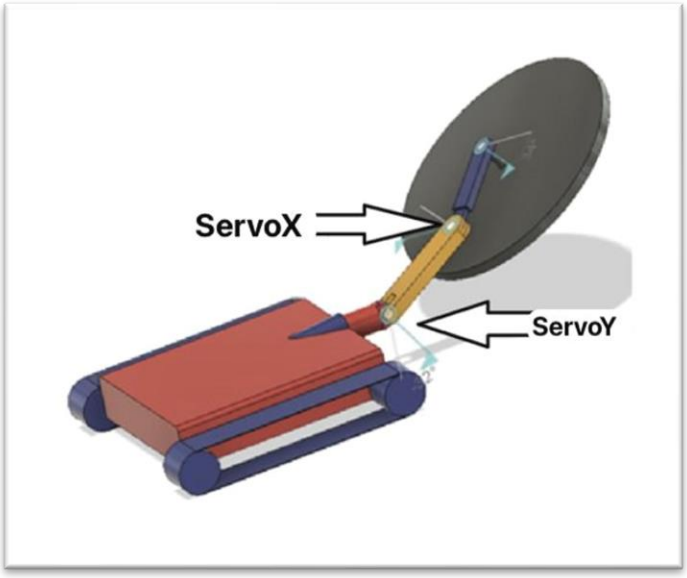


Figure 28: Servos locations in the manipulator.

3.3.4.1 ServoY

ServoY is the first servo motor in the arm and, as mentioned earlier, is responsible for the vertical movement of the entire arm (up/down). During the cleaning process, the arm is positioned downward, while it is raised when the robot is not cleaning.

The worst-case scenario for ServoY occurs when it is operating with no inclination, the angle is 0°.

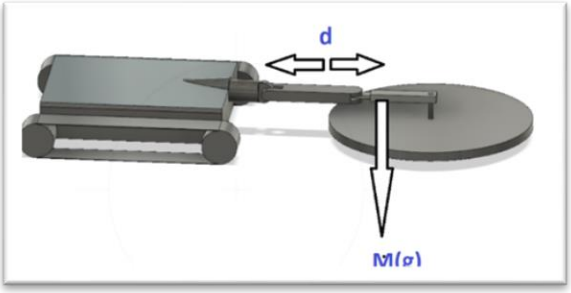


Figure 29: worst-case scenario for ServoY.

To estimate the maximum torque required from the motor, the following equation (3.20) is used: $\tau = f \cdot d$

Where:

- τ is the torque.
- f is the force applied, calculated as $f = M \cdot g$, and g is the acceleration due to gravity, 9.81 m/s^2.
- M is the total mass of the arm (including links, ServoX, brush motor, and the brush), which is approximately 0.6 kg,
- d is the distance from the motor shaft to the center of mass of the arm, estimated to be 15 cm.

Thus, the torque is:

$$\tau = f \cdot d = 0.883\ N.m$$

which is equivalent to 9 kg·cm. And by multiplying this torque by a safety factor of 1.5, the torque required from the motor is 13.5 kg.cm.

As a result, the required torque is 13.5 kg·cm, while the motor chosen for the project has a rated torque of 35 kg·cm. Clearly, the motor is oversized for the application. However, the motor was discounted, and its price was lower than that of the required motor. Therefore, oversizing is acceptable in this case.

A "FEETECH 35KG Waterproof 7.4V 180-Degree High-Speed Metal Gear Digital Servo Motor" was selected to perform this task. The selection was made based on considerations such as cost, motor sizing, and robustness. Additionally, the motor's IP54 waterproof protection feature is essential for its operation in the PanelGuard robot.



Figure 30: ServoY, FEETECH 35KG.

FT5330M servo motor is composed of a sturdy PA aluminum middle shell, a strong steel metal gear, and a hard iron shaft, provide higher stability and provide longer service life.

3.3.4.2 ServoX

ServoY is the second servo motor in the arm and is responsible for the horizontal movement of the second link of the arm. During the cleaning process, ServoX may move the link left and right, or it may remain stationary. However, if the robot is not cleaning, ServoY will not move and will remain in its neutral position.

The worst-case scenario for ServoX occurs when it is operating on an inclined surface at a 45-degree angle.

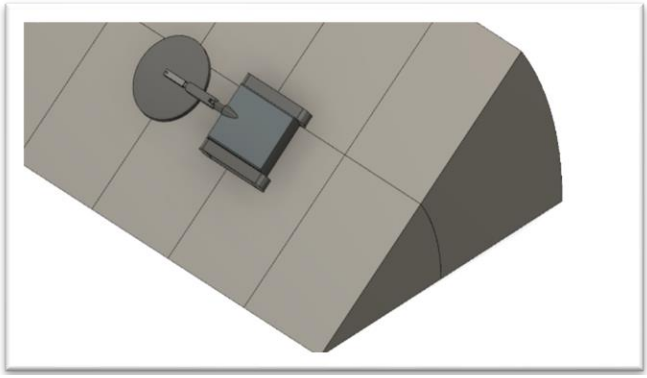


Figure 31: Robot operating on an inclined surface.

To estimate the maximum torque required from the motor, the following equation (3.20) is used: $\tau = f \cdot d$

Where:

- τ is the torque.
- f is the force applied multiplied by sin(45), calculated as $f = M\text{gsin}(45)$, and g is the acceleration due to gravity, 9.81 m/s^2.
- M is the total mass of the second link, brush motor, and brush, which is 0.5 kg.
- d is the distance from the motor shaft to the center of mass of the second link and the brush, estimated to be 10 cm.

Thus, the torque is:

$$\tau = M\text{gsin}(45) \cdot d = 0.35\ N.m$$

which is equivalent 3.7 kg.cm. And by multiplying this torque by a safety factor of 2, the torque required from the motor is 7.4 kg.cm

As a result, the required torque is 7.4 kg·cm, while the motor chosen for the project has a rated torque of 13 kg·cm, which is sufficient.

A "MG995 servo Motor" was selected to perform this task. The selection was made based on considerations such as cost, motor sizing, and quality. Additionally, the motor's waterproof feature is crucial for its operation in the PanelGuard robot.



Figure 32: ServoX, MG995 servo Motor.

3.3.4.3 Brush motor

The brush motor is responsible for rotating the brush to clean the panel. A JSUMO was selected to perform this task.



Figure 33: JSUMO Titan DC Gearhead Motor.

The JSUMO Titan DC Gearhead Motor 12V 500 RPM SP is a versatile geared DC motor, well-suited for applications that require a balance between speed and torque. Operating on a 12V power supply, it provides a no-load speed of 500 RPM. The motor features a robust construction and an integrated gearbox, ensuring reliable performance and efficient power transmission. Its compact design makes it ideal for use in the PANELGUARD system.

- Voltage: 12V
- No Load Speed: 500 RPM
- Shaft Diameter: 6mm
- Motor Dimensions: 37mm x 70mm

3.3.5 sensors

Sensors play an indispensable role in the functionality of a mobile robot. They gather data from the environment, which is then processed by the control system to make informed decisions. The choice of sensors can significantly influence the robot's capabilities [27].

3.3.4.1 Sensors used in the project

- Proximity Sensors

These are used to detect the presence of objects within a certain range. They are crucial for collision avoidance and are commonly used in applications like automated vacuum cleaners and warehouse robots. And in PanelGaurd particularly Infrared and ultrasonic sensors are used to enable the robot to detect edges in addition to any unpredictable obstacles ensuring safe and efficient navigation.

- IR Sensor: The IR (Infrared) sensor enables the detection/monitoring of objects or surfaces based on infrared radiation. The sensor used infrared light to measure distances, detect obstacles, and identify changes in the environment which provides essential feedback for navigation.

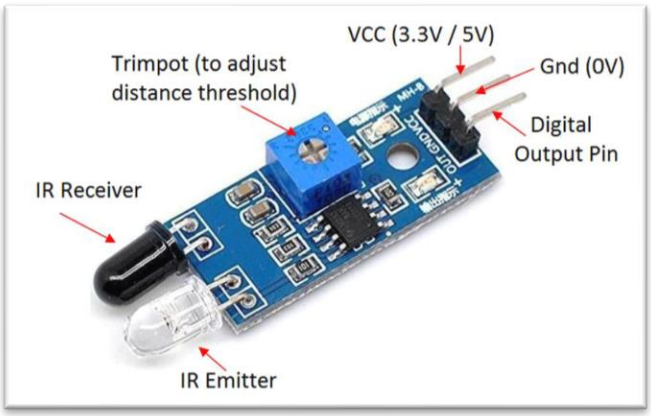


Figure 34: IR sensor module.

It is used for multiple purposes in PanelGaurd Robotic System.

1. To detect the edges of solar panels for emergency stops.
2. To ensure the manipulator is in contact with the surface of the solar panels before starting its operation.

- Ultrasonic Sensor: The ultrasonic sensor enables precise and accurate detection of the solar panel's surface and surrounding environment. The sensor emits high-frequency sound waves and measures the time it takes for the waves to bounce back after hitting an object allowing for precise distance calculation for proper navigation of the robotic system. The ultrasonic sensor used for the robot is the HC-SR04.

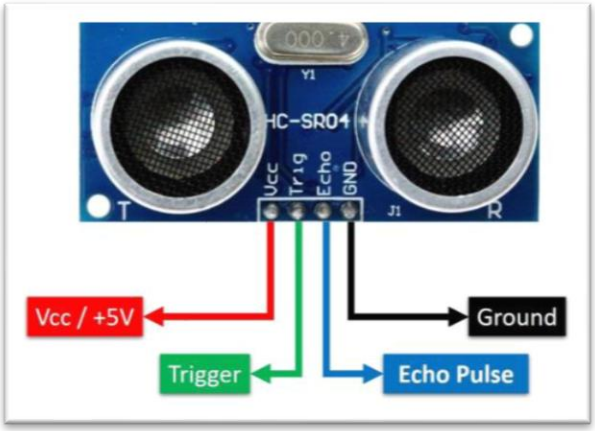


Figure 35: Pin labelling of Ultrasonic sensor module.

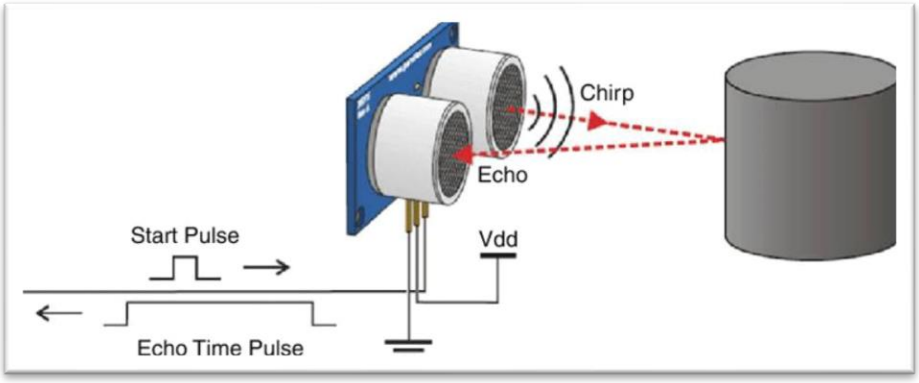


Figure 36: Working principle of ultrasonic sensor.

- Gyroscopes and Accelerometers

These sensors are vital for maintaining the orientation and balance of the robot. Commonly found in bipedal robots and drones [16], they are essential for detecting the incline angle and adjusting motor power accordingly.

3.3.6 Power Supply

The power supply is a critical component that determines how long and how far a robot can operate. It's essentially the lifeblood of the machine.

In the PanelGuard, a power supply system comprising eight batteries was utilized to deliver the necessary electrical energy to the five motors and the ESP microcontroller, ensuring reliable and efficient operation.

The robot is powered by EVE ICR18650-26V 2550mAh 5C 18650 Li-ion batteries, selected after careful consideration of factors such as cost, reliability, and technical specifications.



Figure 37: EVE ICR18650-26V 2550mAh 5C 18650 Li-ion battery.

A features table of the battery is provided below:

Table 9: features table of the battery.

Weight - g	46
Brand	EVE
Model	ICR18650-26V
Size	18650
Battery chemistry	Li-ion
Voltage	3.7V
Min. capacity - mAh	2,500.00
Typ. capacity - mAh	2,550.00
Battery version	Flat top
Discharge current - A	7.65
Circuit protection	Unprotected
Height - mm	65.10
Diameter - mm	18.20

3.3.6.1 Power supply Connections:

The configuration consists of eight batteries, with six connected in series to provide approximately 24 V, and an additional series connection of two batteries to provide 7.4V.

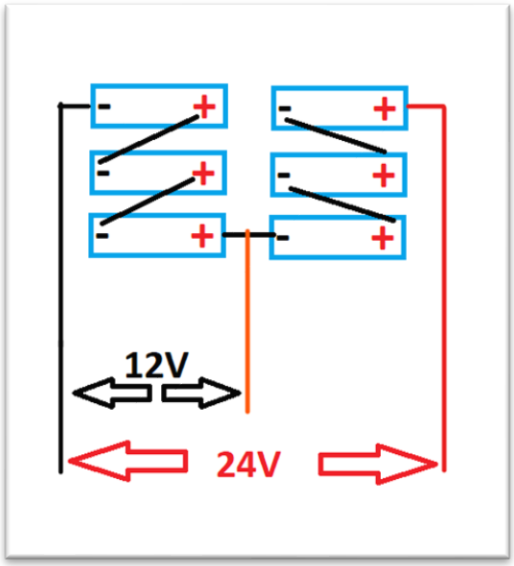


Figure 38: The connection of six batteries in series.

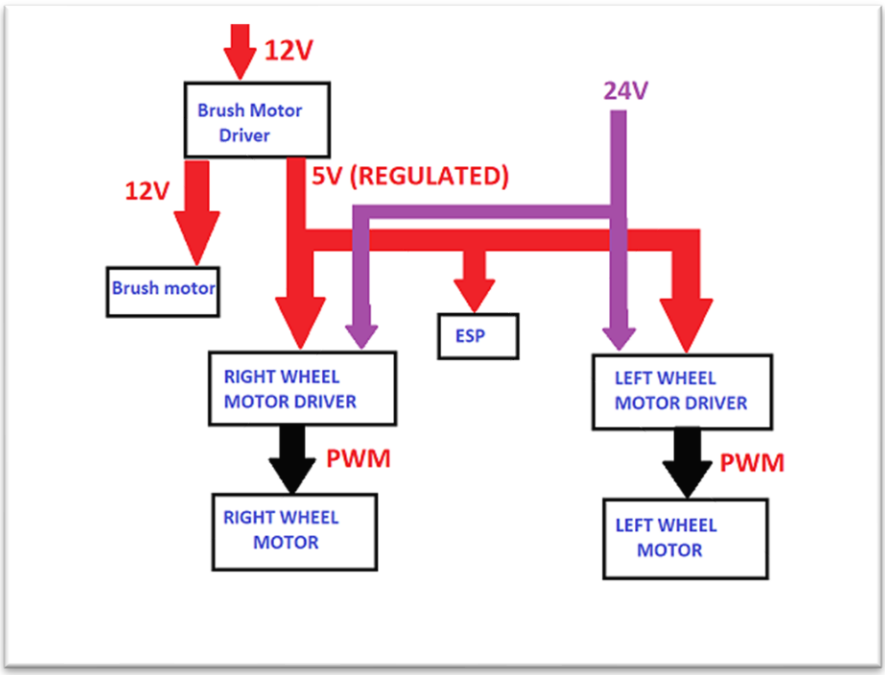


Figure 39 : Power diagram.

Two batteries are connected in series to directly supply power to two servo motors without the use of a motor driver.

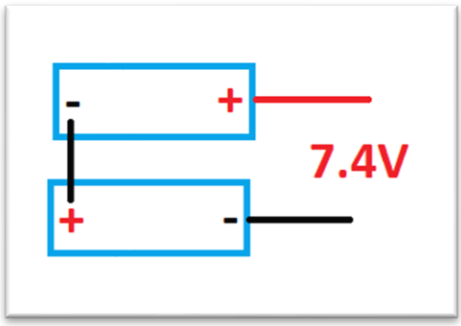


Figure 40: The connection of two batteries in series.

The positive terminal is connected to the VCC pins of the servos, and the negative terminal is part of the ground network.

3.4 Software, and Coding Implementation



Figure 41: Arduino ide interface.

This code sets up the hardware connections and prepares the necessary components for controlling motors and servos using an ESP32 microcontroller.

3.4.1 flow chart

For Clearer idea, the robot's operational flowchart is divided into two distinct branches: one for remote control and another for autonomous operation (including sensor data processing, edge detection, and obstacle avoidance). These branches are interconnected via two circular symbols representing the 'Remote Control' and 'Running'.

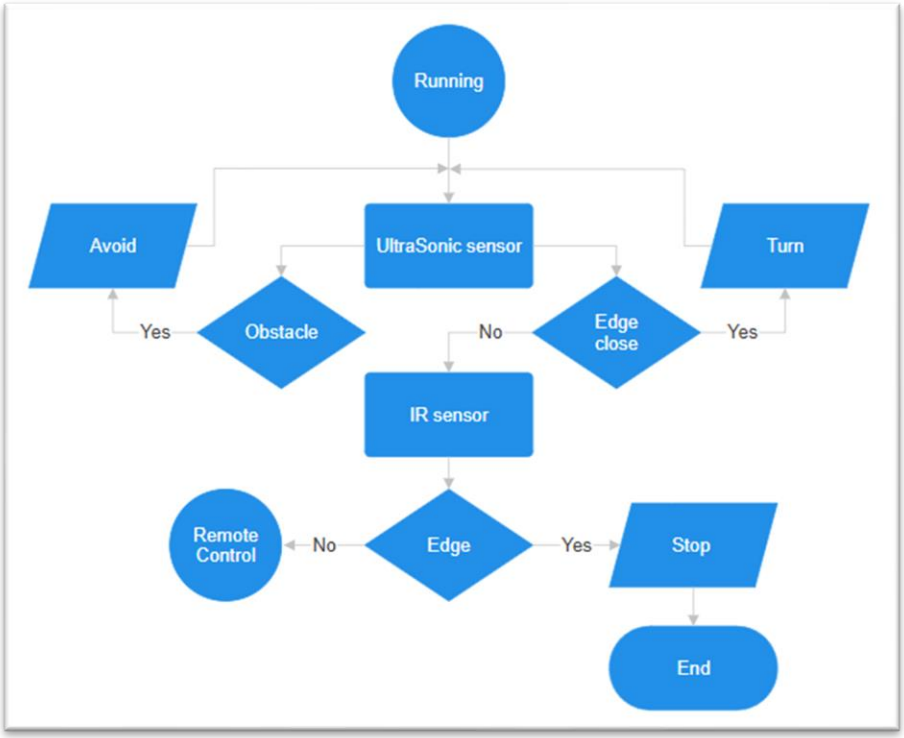


Figure 42: Flow chart (1), Autonomous operation.

While the robot is in 'Running' state, the Ultrasonic Sensor continuously scans for obstacles and also checks if the solar panel's edge is close. If an obstacle is detected, the robot initiates an obstacle avoidance maneuver. If no obstacle is detected, the robot proceeds. To enhance safety, an IR sensor is used to double-check for proximity to the edge. If the IR sensor detects the robot is on the edge, an emergency stop is triggered.

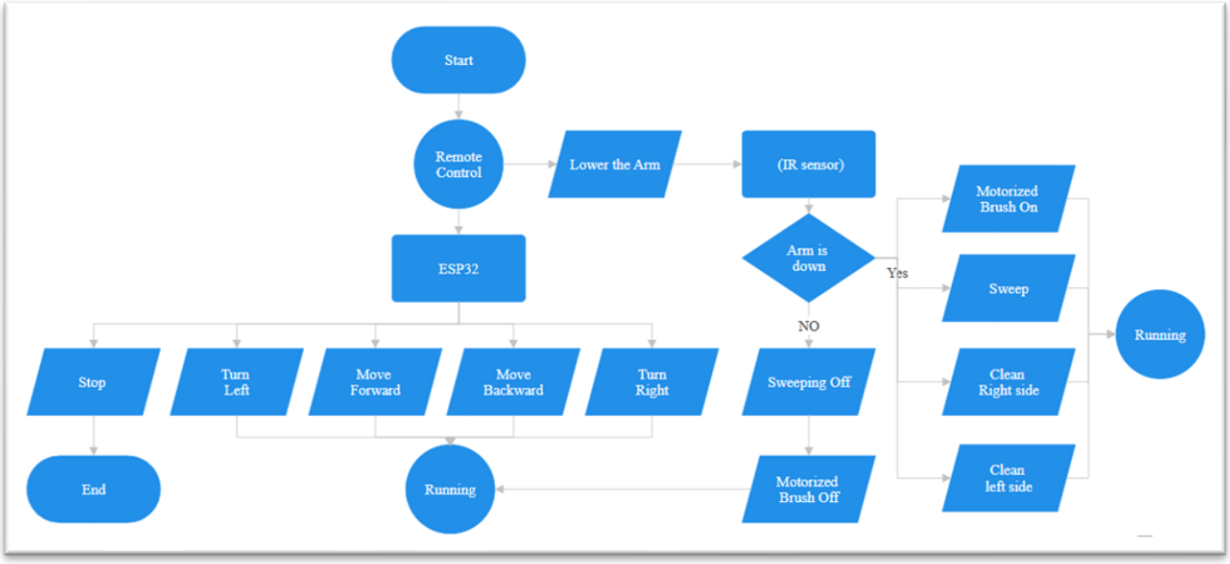


Figure 43: Flow chart (2), Remote Control.

When the robot begins its operation. The robot continuously monitors for input from the remote control. Based on the received input, it executes actions such as turning left, turning right, moving forward, moving backward, or stopping.

For arm control, based on a certain input the robot will lower the arm. But To enhance safety, an IR sensor verifies successful arm lowering. If the arm is successfully lowered, the robot can activate the motorized brush and sweep or position the arm to the right or left. If the arm fails to lower, no arm-related actions (cleaning, positioning) are permitted until successful lowering is achieved.

Chapter 4: Mathematical model and Analysis

While analyzing the dynamic characteristics of PanelGaurd robot, the highly present similarities between the system and the double inverted pendulum showed that the optimal choice for the modeling is the Lagrange approach [28], since it's a common method for modeling inverted pendulums.

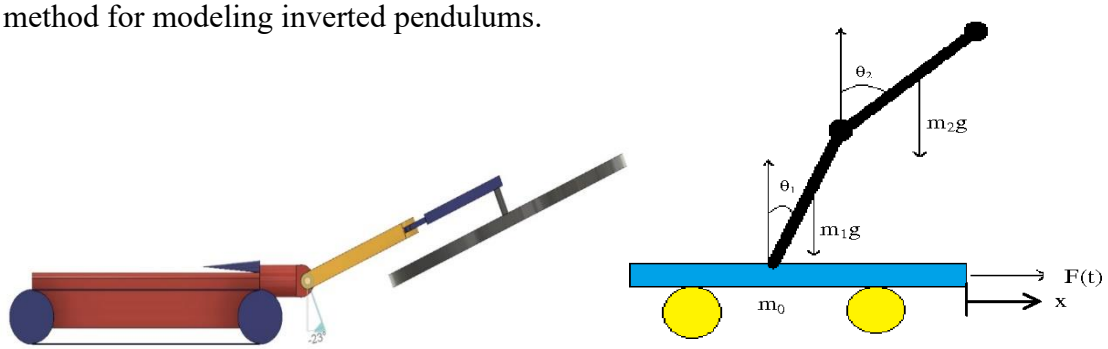


Figure 44: Similarity between the robot and the double inverted pendulum.

Different from the Newtonian mechanics that is based on forces, the Lagrangian method is based on the energies of the system [29]. The Lagrangian L is defined as the difference between the kinetic energy KE and the potential energy PE of the system.

$$L = KE - PE \tag{4.1}$$

This approach is particularly advantageous when dealing with systems where the forces are not easily characterized by simple vectors, or in systems with constraints.

Through the Euler-Lagrange the differential equations are:

$$\frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i \tag{4.2}$$

where $q_i, \dot{q}_i$ are the generalized coordinates and their time derivatives respectively, and Q_i is the external force/torque applied to joint i.

The equations of motion can be expressed as the following:

$$M(q)\ddot{q} + C(q,\dot{q}) + G(q) = \tau \tag{4.3}$$

Where $M(q)\ddot{q}$ is the inertia matrix, it's a square matrix that depends on the robot's configuration and captures how the mass distribution affects its motion. And $C(q,\dot{q})$ is the Coriolis and centrifugal matrix, which also depends on the robot's configuration and velocities. G(q) represents the force/torque due to gravity acting on the robot. And τ is the Joint Torque/Force Term, this is the vector of joint torques, or forces applied to the robot to control its motion [28].

4.1 Manipulator with respect to cart model

4.1.1 Transformation matrices of the manipulator

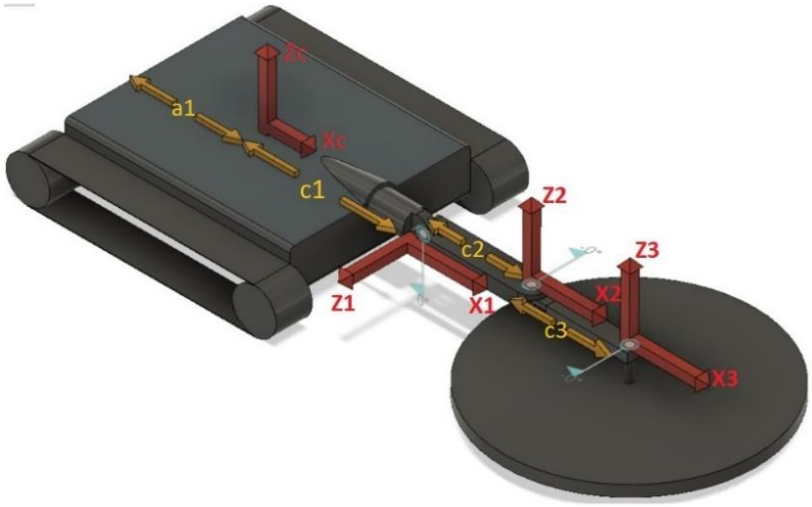


Figure 45: DH manipulator frames.

The frames representing the manipulator are shown in figure 5, and the DH frames are used to find DH parameters, to also find the transformation matrices. Assuming the cart frame is fixed.

Table 10: DH parameters for the manipulator.

link	$\boldsymbol{\theta}$	$\boldsymbol{\alpha}$	$\boldsymbol{a}$	$\boldsymbol{d}$
1	0	90^0	C_1	0
2	$\theta_1(t)$	-90^0	C_2	0
e	$\theta_2(t)$	0	C_3	0

The DH parameter is a set of four variables that aid in the mathematical representation of each joint [28].

- $\boldsymbol{a_i}$ = distance from $\boldsymbol{Z_{i-1}}$ to $\boldsymbol{Z_i}$ measured along $\boldsymbol{X_i}$
- $\boldsymbol{\alpha_i}$ = angle measured around $\boldsymbol{X_i}$ from $\boldsymbol{Z_{i-1}}$ to $\boldsymbol{Z_i}$
- $\boldsymbol{d_i}$ = distance between $\boldsymbol{X_{i-1}}$ and $\boldsymbol{X_i}$ as measured along $\boldsymbol{Z_{i-1}}$
- $\boldsymbol{\theta_i}$ = angle measured around $\boldsymbol{Z_{i-1}}$ from $\boldsymbol{X_{i-1}}$ to $\boldsymbol{X_i}$

The transformation for each connection can be written using the DH parameters by (4.4):

$$T_i^{i-1} = \begin{matrix} \cos(\theta_i) & -\sin(\theta_i)\cos(\alpha_i) & \sin(\theta_i)\sin(\alpha_i) & a_n\cos(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i)\cos(\alpha_i) & -\cos(\theta_i)\sin(\alpha_i) & a_n\sin(\theta_i) \\ 0 & \sin(\alpha_i) & \cos(\alpha_i) & d_n \\ 0 & 0 & 0 & 1 \end{matrix}$$

(4.5 - 4.8) respectively:

$$T_1^{M_C} = \begin{bmatrix} 1 & 0 & 0 & C_1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad T_{m_1}^1 = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & C_{m_1}\cos\theta_1 \\ \sin\theta_1 & \cos\theta_1 & 0 & C_{m_1}\sin\theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2^1 = \begin{bmatrix} \cos\theta_1 & 0 & -\sin\theta_1 & C_2\cos\theta_1 \\ \sin\theta_1 & 0 & \cos\theta_1 & C_2\sin\theta_1 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} T_e^2 = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & C_3\cos\theta_2 \\ \sin\theta_2 & \cos\theta_2 & 0 & C_3\sin\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Where end effector frame e, is m_2 frame as well.

$$P_{m_1}^1 = T_{m_1}^1(1{:}3,4) = \begin{bmatrix} C_{m_1}\cos\theta_1 \\ C_{m_1}\sin\theta_1 \\ 0 \end{bmatrix} \tag{4.9}$$

This represents the position of link 1 mass (m_1) in frame 1, extracted from the transformation matrix $T_{m_1}^1$.

$$P_{m_2}^1 = T_e^1\ (1{:}3,4) = \begin{bmatrix} c_2cos\theta_1 + c_3cos\theta_1cos\theta_2 \\ c_2sin\theta 1 + c_3sin\theta_1cos\theta_2 \\ -C_3sin\theta_2 \end{bmatrix} \tag{4.10}$$

And this represents the position of the end-effector (e) in frame 1, extracted from the transformation matrix $T_e^1 = T_2^1 T_e^2$.

4.1.2 Manipulator's Jacobians

Jacobian of mass 1 in frame 1 with respect to $q_{m_1} = [\theta_1]$ **.**

$$J_{m_1} = \begin{matrix} \frac{\partial P_{m_1}^1(1)}{\partial\theta_1} \\ \frac{\partial P_{m_1}^1(2)}{\partial\theta_1} \end{matrix} \tag{4.11}$$

Jacobian of end-effector in frame 1 with respect to $q_{m_2} = [\ \theta_1 \quad \theta_2\ ]$.

$$J_e = \begin{matrix} \frac{\partial P_{m_2}^1(1)}{\partial\theta_1} & \frac{\partial P_{m_2}^1(1)}{\partial\theta_2} \\ \frac{\partial P_{m_2}^1(2)}{\partial\theta_1} & \frac{\partial P_{m_2}^1(2)}{\partial\theta_2} \\ \frac{\partial P_{m_2}^1(3)}{\partial\theta_1} & \frac{\partial P_{m_2}^1(3)}{\partial\theta_2} \end{matrix} \tag{4.12}$$

The kinetic energy and that the potential energy to obtain the dynamical equations are:

$$KE_i = 0.5\sum_{i=1}^{2} m_i \cdot (\ \dot{q_{m_i}}^T . J_i^T \cdot J_i \cdot \dot{q_{m_1}}) \tag{4.13}$$

$$PE_i = \sum_{i=1}^{2} m_i\ .\ g\ .\ H_i \tag{4.14}$$

Applying Lagrangian, (4.2):

$$\tau_1 = \frac{d}{dt}\left(\frac{\partial \mathrm{L}}{\partial \dot{\theta_1}}\right) - \frac{\partial L}{\partial \theta_1}$$

$$\tau_2 = \frac{d}{dt}\left(\frac{\partial \mathrm{L}}{\partial \dot{\theta_2}}\right) - \frac{\partial \mathrm{L}}{\partial \theta_2}$$

Where the external forces should be applied to the joints are τ_1, τ_2 from the motors.

So, the equation that represents the dynamics of a robotic manipulator is:

$$\mathrm{M}(q_i)\ddot{q_i} + C(q_i, \dot{q}_i) + Nq_i^2 + G(q_i) = \tau_i \qquad (4.3)$$

$$M = \begin{bmatrix} m_1 C_{m_1}^2 + m_2 c_2^2 + 2m_2 c_2 c_3 \cos(\theta_2(t)) + m_2 c_3^2 cos^2(\theta_2(t)) & 0 \\ 0 & m_2 c_3^2 \end{bmatrix}$$

$$\ddot{q} = \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix}$$

$$N = \begin{bmatrix} 0 & 0 \\ \frac{m_2 \sin(2\theta_2(t)) c_3^2}{2} + c_2 m_2 \sin(\theta_2(t)) c_3 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta_1}^2 \\ \dot{\theta_2}^2 \end{bmatrix}$$

$$G = \begin{bmatrix} C_{m_1} m_1 \cos(\theta_1(t)) + c_2 m_2 \cos(\theta_1(t)) + c_3 m_2 \cos(\theta_1(t)) \cos(\theta_2(t)) \\ -c_3 m_2 \sin(\theta_1(t)) \sin(\theta_2(t)) \end{bmatrix} g$$

$$C = \begin{bmatrix} -m_2 {c_3}^2 \sin(2\theta_2(t)) - 2c_2 c_3 m_2 \sin(\theta_2(t)) \\ 0 \end{bmatrix} [\dot{\theta}_1 \dot{\theta}_2]$$

$$\tau = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix}$$

Table 11 Parameters of Manipulator with respect to cart model

Parameter:	Description:
$\theta_1(t)$	The angular position of the first joint as a function of time.
$\theta_2(t)$	The angular position of the second joint as a function of time.
m_1	The mass of the first link.
m_2	The mass of the second link (end effector).
C_{m_1}	Distance Z_c to the center of mass of link 1.
c_1	Distance from Z_c to Z_1 (center of mass to the first joint).
c_2	Distance from Z_1 to Z_2 (first link length).
c_3	Distance from Z_2 to Z_3 (second link length).
τ_1	represent the torque applied at the first joint.
τ_2	represent the torque applied at the second joint.

4.2 Cart, base and ground model:

4.2.1 Cart, base transformation matrix

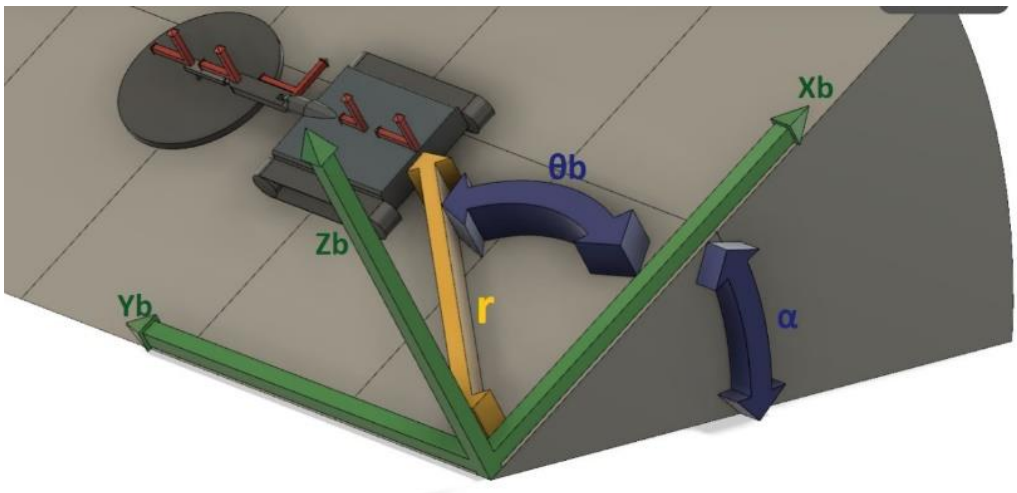


Figure 46: polar coordinates.

The transformation matrix T_c^b that describes the motion of the cart relative to the base and its position represented by polar coordinates $(r(t), \theta_b)$ is given by:

$$T_c^b = \begin{bmatrix} \cos\theta_b & -\sin\theta_b & 0 & r(t)\cos\theta_b \\ \sin\theta_b & \cos\theta_b & 0 & r(t)\sin\theta_b \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.15}$$

The other matrix transfer's the center of mass frame to frame of rotation in the edge of the cart:

$$T_{M_c}^c = \begin{bmatrix} \cos\theta_s & -\sin\theta_s & 0 & a_1\cos\theta_s \\ \sin\theta_s & \cos\theta_s & 0 & a_1\sin\theta_s \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.16}$$

Multiplying these two matrices gives the transformation matrix that describes the cart position and orientation with respect to the base frame which is the solar panel:

$$T_{M_c}^b = T_c^b . T_{M_c}^c \tag{4.17}$$

Taking the position from $T_{M_c}^b$ matrix:

$$P_{M_c}^b = T_{M_c}^b(1{:}3,4) = \begin{bmatrix} a_1\cos(\theta_b(t) + \theta_s(t)) + \cos(\theta_b(t))r(t) \\ a_1\sin(\theta_b(t) + \theta_s(t)) + \sin(\theta_b(t))r(t) \\ 0 \end{bmatrix} \tag{4.18}$$

4.2.2 Cart to base Jacobian matrix:

The Jacobian matrix of the Cart coordinates in terms of the partial derivatives of two functions $P_{M_c}^b(1)$ and $P_{M_c}^b(2)$ with respect to $\theta_b(t)$, $r(t)$, and $\theta_s(t)$.

$$J_{v_c} = \begin{matrix} \frac{\partial P_{M_c}^b(1)}{\partial\theta_b} & \frac{\partial P_{M_c}^b(1)}{\partial r} & \frac{\partial P_{M_c}^b(1)}{\partial\theta_s} \\ \frac{\partial P_{M_c}^b(2)}{\partial\theta_b} & \frac{\partial P_{M_c}^b(2)}{\partial r} & \frac{\partial P_{M_c}^b(2)}{\partial\theta_s} \end{matrix} \tag{4.19}$$

This is the Jacobian matrix of the Cart:

$$J_{v_c} = \begin{matrix} -a_1\sin(\theta_b(t)+\theta_s(t)) - \sin(\theta_b(t))r(t) & \cos(\theta_b(t)) & -a1\sin(\theta_b(t)+\theta_s(t)) \\ a_1\cos(\theta_b(t)+\theta_s(t)) + \cos(\theta_b(t))r(t) & \sin(\theta_b(t)) & a1\cos(\theta_b(t)+\theta_s(t)) \end{matrix}$$

4.2.3 Base, ground transformation matrix:

Given the inclined panel in which the robot is operating, a rotational frame was implemented in order to define the inclined surface in relation to the ground.

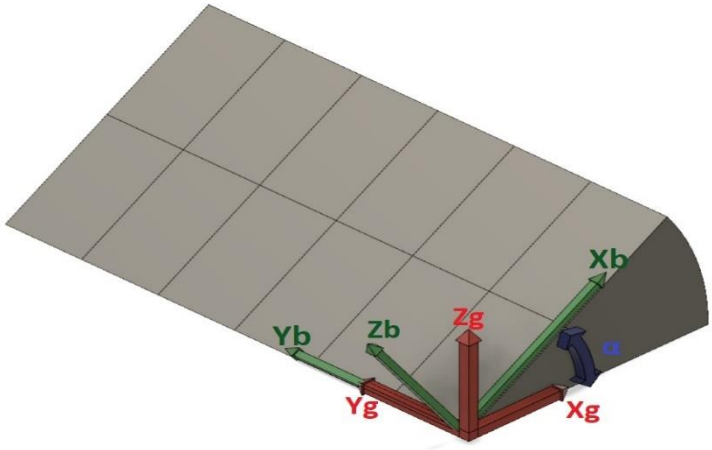


Figure 47 rotational frame of the panel

As it's clear in the figure there is no translation involved. and assuming the worst-case scenario for the robot to operate, the panel is inclined at a 45-degree angle around y-axis [30]. Therefore, the transformation matrix that represents this inclined panel with respect to the ground is:

$$T_b^g = \begin{bmatrix} \cos\frac{\pi}{4} & 0 & -\sin\frac{\pi}{4} & 0 \\ 0 & 1 & 0 & 0 \\ \sin\frac{\pi}{4} & 0 & \cos\frac{\pi}{4} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4.20}$$

Height of the center of mass from the ground**:**

$$H_{M_c} = T_{M_c}^g(3,4) == \frac{\sqrt{2}a_1\cos(\theta_b(t)+\theta_s(t))}{2} + \frac{\sqrt{2}\cos(\theta_b(t))r(t)}{2} \tag{4.21}$$

This represents the height of the center of mass from the ground, extracted from the transformation matrix $T_{M_c}^g = T_b^g . T_c^b . T_{M_c}^c$

Kinetic energy of the cart mass (M_c)**:**

$$KE_{M_c} = \frac{1}{2} M_c (\dot{q_{M_c}}^T . J_{v_c}^T \cdot J_{v_c} \cdot \dot{\mathrm{q}_{\mathrm{M_c}}}) + \frac{1}{2} I_c \dot{\theta_s}^2 \quad (4.22)$$

This is the kinetic energy of mass (Mc), where q_{M_c} the generalized coordinates

$$q_{M_c} = [\, \theta_b \quad r(t) \quad \theta_s]$$

Potential energy of the cart mass (M_c)**:**

$$PE_{M_c} = M_c . g . H_{M_c} \quad (4.23)$$

J_{M_c} is the Jacobian matrix of the Cart coordinates, I_c is the moment of inertia, and θ_s is the steering angle of the cart.

$$L_c = KE_{M_c} - PE_{M_c} \quad (4.1)$$

Applying Lagrangian, which is the difference between the kinetic energy (KE_{M_c}) and the potential energy (PE_{M_c}).

Then through the Euler-Lagrange (4.2) the differential equations are:

$$0 = \frac{d}{dt}\left(\frac{\partial L_c}{\partial \dot{\theta_b}}\right) - \frac{\partial L_c}{\partial \theta_b}$$

$$u_1 + u_2 = \frac{d}{dt}\left(\frac{\partial L_c}{\partial \dot{r}}\right) - \frac{\partial L_c}{\partial r}$$

$$\frac{L}{2} u_1 - \frac{L}{2} u_2 = \frac{d}{dt}\left(\frac{\partial L_c}{\partial \dot{\theta_s}}\right) - \frac{\partial L_c}{\partial \theta_s}$$

Where the external forces considered $u_1,\ u_2$ are the control inputs to the two motors that drive the robot's wheels.

$$\mathrm{M}(q_i)\ddot{q}_\iota + C(q_i, \dot{q}_i) + N(q_i)q_i^2 + G(q_i) = \tau_i \tag{4.3}$$

$$M = \begin{bmatrix} M_c a_1^2 + 2M_c \cos(\theta_s) a_1 r(t) + M_c r(t)^2 & -M_c a_1 \sin(\theta_s) & M_c a_1^2 + M_c \cos(\theta_s)\, r(t) a_1 \\ -M_c a_1 \sin(\theta_s) & M_c & -M_c a_1 \sin(\theta_s) \\ M_c a_1^2 + M_c \cos(\theta_s)\, r(t) a_1 & -M_c a_1 \sin(\theta_s) & M_c a_1^2 + I_c \end{bmatrix}$$

$$\ddot{q} = \begin{bmatrix} \ddot{\theta_b} \\ \ddot{r} \\ \ddot{\theta_s} \end{bmatrix}$$

$$N = \begin{bmatrix} 0 & 0 & -M_c a_1 \sin(\theta_s(t)) r(t) \\ -M_c r(t) - M_c a_1 \cos(\theta_s(t)) & 0 & -M_c a_1 \cos(\theta_s(t)) \\ M_c a_1 \sin(\theta_s(t)) r(t) & 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta_b}^2 \\ \dot{r}^2 \\ \dot{\theta_s}^2 \end{bmatrix}$$

$$G = \begin{bmatrix} -M_c(a_1 \sin(\theta_b(t) + \theta_s(t)) + \sin(\theta_b(t)) r(t) \\ M_c \cos(\theta_b(t)) \\ -M_c a_1 \sin(\theta_b(t) + \theta_s(t)) \end{bmatrix} \frac{g\sqrt{2}}{2}$$

$$C = \begin{bmatrix} 2M_c r(t) + 2M_c a_1 \cos(\theta_s(t)) & -2M_c a_1 \sin(\theta_s(t))\, r(t) & 0 \\ 0 & -2M_c a_1 \cos(\theta_s(t)) & 0 \\ 2M_c a_1 \cos(\theta_s(t)) & 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta_b} \dot{r} \\ \dot{\theta_b} \dot{\theta_s} \\ \dot{r}\, \dot{\theta_s} \end{bmatrix}$$

$$\tau = \begin{bmatrix} 0 \\ u_1 + u_2 \\ \frac{L}{2} u_1 - \frac{L}{2} u_2 \end{bmatrix}$$

Table 12 Parameters of cart model

Parameter:	Description:
$\theta_b(t)$	The angle measured from X_b to $r(t)$ as a function of time.
$\theta_s(t)$	The steering angle as a function of time.
$r(t)$	Distance from base frame to the first frame of the cart.
a_1	Constant distance between the two frames of the cart.
M_c	The mass of cart.
I_c	This is the inertia mass of the cart.
u_1	Represent the force applied at the first motor. (input)
u_2	Represent the torque applied at the second motor. (input)

4.3 Integrated Robotic System Model

The integrated model enables the observation and analysis of the system's behavior as a whole, rather than individual parts. It offers a deeper understanding of how different components interact and work together to enhance the robot's overall functionality [31]. This section aims to delve into the complexities of the integrated system in order to enhance its performance and efficiency.

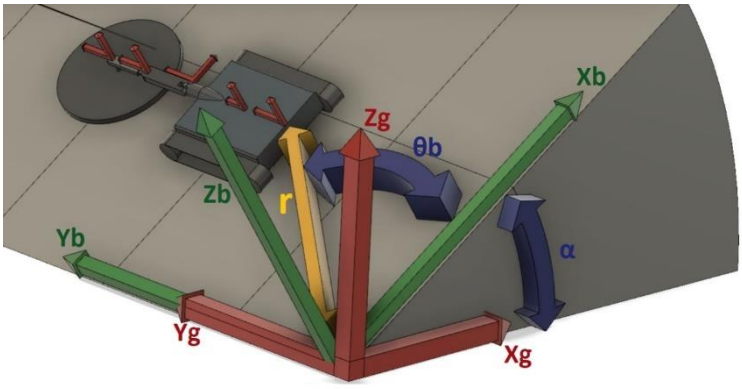


Figure 48 all frames for the system

Now after T_b^g, T_c^b, T_1^c, T_2^1, and T_e^2 were found, (Figure 8) shows all the frames assigned for the robot for better understanding.

Using Lagrange, the kinetic and potential energies:

$$KE_i = 0.5 \sum_{i=1}^{5} M_i \cdot (\dot{q_i}^T . J_i^T \cdot J_i \cdot \dot{q}_i) \qquad (4.24)$$

$$PE_i = \sum_{i=1}^{5} M_i \, . \, g \, . \, H_i$$

4.3.1 Jacobians for the whole system

For Link 1:

Jacobian matrix J_1 of link 1 in terms of the partial derivatives of $P_{m_1}^b = T_{m_1}^b(1{:}3,4)$ with respect to the generalized coordinates $q = [\theta_b \quad r(t) \quad \theta_s \quad \theta_1]$

$$J_{m_1}^b = \begin{matrix} \frac{\partial P_{m_1}^b(1)}{\partial \theta_b} & \frac{\partial P_{m_1}^b(1)}{\partial r} & \frac{\partial P_{m_1}^b(1)}{\partial \theta_s} & \frac{\partial P_{m_1}^b(1)}{\partial \theta 1} \\ \frac{\partial P_{m_1}^b(2)}{\partial \theta_b} & \frac{\partial P_{m_1}^b(2)}{\partial r} & \frac{\partial P_{m_1}^b(2)}{\partial \theta_s} & \frac{\partial P_{m_1}^b(2)}{\partial \theta 1} \\ \frac{\partial P_{m_1}^b(3)}{\partial \theta_b} & \frac{\partial P_{m_1}^b(3)}{\partial r} & \frac{\partial P_{m_1}^b(3)}{\partial \theta_s} & \frac{\partial P_{m_1}^b(3)}{\partial \theta 1} \end{matrix} \tag{4.26}$$

For Link 2:

Jacobian matrix J_e^b of link 1 in terms of the partial derivatives of $P_{m_2}^b = T_e^b(1{:}3,4)$ with respect to the generalized coordinates $q = [\theta_b \quad r(t) \quad \theta_s \quad \theta_1 \quad \theta_2\ ]$

$$J_e^b = \begin{matrix} \frac{\partial P_{m_2}^b(1)}{\partial \theta_b} & \frac{\partial\, P_{m_2}^b 1)}{\partial r} & \frac{\partial\, P_{m_2}^b(1)}{\partial \theta_s} & \frac{\partial\, P_{m_2}^b(1)}{\partial \theta_1} & \frac{\partial\, P_{m_2}^b(1)}{\partial \theta_2} \\ \frac{\partial P_{m_2}^b(2)}{\partial \theta_b} & \frac{\partial\, P_{m_2}^b(2)}{\partial r} & \frac{\partial\, P_{m_2}^b(2)}{\partial \theta_s} & \frac{\partial\, P_{m_2}^b(2)}{\partial \theta_1} & \frac{\partial\, P_{m_2}^b(2)}{\partial \theta_2} \\ \frac{\partial P_{m_2}^b(3)}{\partial \theta_b} & \frac{\partial\, P_{m_2}^b(3)}{\partial r} & \frac{\partial\, P_{m_2}^b(3)}{\partial \theta_s} & \frac{\partial\, P_{m_2}^b(3)}{\partial \theta_1} & \frac{\partial\, P_{m_2}^b(3)}{\partial \theta_2} \end{matrix} \tag{4.27}$$

For the potential energies, the height of each link with respect to the ground will be extracted as: (4.28 – 4.31) respectively:

$$H_{m1} = T_{m_1}^g(3,4)$$

$$H_{m2} = T_{m_2}^g(3,4)$$

$$H_{m1} = \frac{\sqrt{2}}{2}(a_1 \cos(\theta_b + \theta_s) + c_1 \cos(\theta_b + \theta_s) + \cos(\theta_b)\, r(t) + C_{m_1}(\cos(\theta_1)\cos(\theta_b + \theta_s) + \sin(\theta_1)))$$

$$H_{m2} = \frac{\sqrt{2}}{2}(c_2 \sin(\theta_1) + c_3 \cos(\theta_2) \sin(\theta_1) + a_1 \cos(\theta_b + \theta_s) + c_1 \cos(\theta_b + \theta_s) + \cos(\theta_b)\, r(t) + \cos(\theta_b + \theta_s)\,(c_2 \cos(\theta_1) + c_3 \cos(\theta_1)\cos(\theta_2)) - c_3 \sin(\theta_2)\sin(\theta_b + \theta_s))$$

For the cart:

Same as in cart modeling section. The Jacobian matrix of the Cart coordinates in terms of the partial derivatives of two functions $P_{M_c}^b(1)$ and $P_{M_c}^b(2)$ with respect to $\theta_b(t)$, $r(t)$, and $\theta_s(t)$.

$$J_{v_c} = \begin{matrix} \frac{\partial P_{M_c}^b(1)}{\partial \theta_b} & \frac{\partial P_{M_c}^b(1)}{\partial r} & \frac{\partial P_{M_c}^b(1)}{\partial \theta_s} \\ \frac{\partial P_{M_c}^b(2)}{\partial \theta_b} & \frac{\partial P_{M_c}^b(2)}{\partial r} & \frac{\partial P_{M_c}^b(2)}{\partial \theta_s} \end{matrix} \qquad (4.32)$$

And the height of the center of mass from the ground for its potential energy is:

$$H_{M_c} = T_{M_c}^g(3,4) == \frac{\sqrt{2} a_1 \cos(\theta_b(t) + \theta_s(t))}{2} + \frac{\sqrt{2}\cos(\theta_b(t)) r(t)}{2} \qquad (4.33)$$

Through the Euler-Lagrange (4.2) the differential equations are:

$$0 = \frac{d}{dt}\left(\frac{\partial \mathrm{L}}{\partial \dot{\theta}_b}\right) - \frac{\partial \mathrm{L}}{\partial \theta_b}$$

$$u_1 + u_2 = \frac{d}{dt}\left(\frac{\partial \mathrm{L}}{\partial \dot{r}}\right) - \frac{\partial \mathrm{L}}{\partial r}$$

$$\frac{L}{2} u_1 - \frac{L}{2} u_2 = \frac{d}{dt}\left(\frac{\partial \mathrm{L}}{\partial \dot{\theta}_s}\right) - \frac{\partial \mathrm{L}}{\partial \theta_s}$$

$$\tau_1 = \frac{d}{dt}\left(\frac{\partial \mathrm{L}}{\partial \dot{\theta}_1}\right) - \frac{\partial \mathrm{L}}{\partial \theta_1}$$

$$\tau_2 = \frac{d}{dt}\left(\frac{\partial \mathrm{L}}{\partial \dot{\theta}_2}\right) - \frac{\partial \mathrm{L}}{\partial \theta_2}$$

And the equations of motion for the whole robotic system can be expressed as the following from equation (4.3):

$$\mathrm{M}(q_i)\ddot{q}_i + C(q_i, \dot{q}_i) + N(q_i^2) + G(q_i) = \tau_i$$

$$\begin{bmatrix} A_1 & B_1 & C_1 & D_1 & E_1 \\ A_2 & B_2 & C_2 & D_2 & E_2 \\ A_3 & B_3 & C_3 & D_3 & E_3 \\ A_4 & B_4 & C_4 & D_4 & E_4 \\ A_5 & B_5 & C_5 & D_5 & E_5 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_b \\ \ddot{r} \\ \ddot{\theta}_s \\ \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} + C(q_i, \dot{q}_i) + N(q_i^2) + G(q_i) = \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \\ \tau_4 \\ \tau_5 \end{bmatrix}$$

The 5x5 inertia matrix describes how the mass distribution affects its motion. Each term is provided below.

The vector representing control input torques is:

$$\tau_i = \begin{bmatrix} 0 \\ u_1 + u_2 \\ \frac{L}{2}u_1 - \frac{L}{2}u_2 \\ \tau_1 \\ \tau_2 \end{bmatrix}$$

Any other term such as Coriolis, Normal and Gravitational forces, will be ignored.

……

$A_1 = r(t)^2(M_c + m_1 + m_2) + r(t)(2M_c a_1 \cos(\theta_s(t)) + 2a_1 m_1 \cos(\theta_s(t)) +$
$2a_1 m_2 \cos(\theta_s(t)) + 2c_1 m_1 \cos(\theta_s(t)) + 2c_1 m_2 \cos(\theta_s(t)) +$
$2C_{m1} m_1 \cos(\theta_1(t)) \cos(\theta_s(t)) + 2c_2 m_2 \cos(\theta_1(t)) \cos(\theta_s(t)) -$
$2c_3 m_2 \sin(\theta_2(t)) \sin(\theta_s(t)) + 2c_3 m_2 \cos(\theta_1(t)) \cos(\theta_2(t)) \cos(\theta_s(t))) + M_c a_1^2 +$
$a_1^2 m_1 + a_1^2 m_2 + c_1^2 m_1 + c_1^2 m_2 + c_3^2 m_2 + C_{m1}^2 m_1 \cos(\theta_1(t))^2 + c_2^2 m_2 \cos(\theta_1(t))^2 -$
$c_3^2 m_2 \cos(\theta_2(t))^2 + 2a_1 c_1 m_1 + 2a_1 c_1 m_2 + 2C_{m1} a_1 m_1 \cos(\theta_1(t)) +$
$2C_{m1} c_1 m_1 \cos(\theta_1(t)) + 2a_1 c_2 m_2 \cos(\theta_1(t)) + 2c_1 c_2 m_2 \cos(\theta_1(t)) +$
$c_3^2 m_2 \cos(\theta_1(t))^2 \cos(\theta_2(t))^2 + 2a_1 c_3 m_2 \cos(\theta_1(t)) \cos(\theta_2(t)) +$
$2c_1 c_3 m_2 \cos(\theta_1(t)) \cos(\theta_2(t)) + 2c_2 c_3 m_2 \cos(\theta_1(t))^2 \cos(\theta_2(t))$

$$B_1 = -M_c a_1 \sin(\theta_s(t)) - a_1 m_1 \sin(\theta_s(t)) - a_1 m_2 \sin(\theta_s(t)) - c_1 m_1 \sin(\theta_s(t)) - c_1 m_2 \sin(\theta_s(t)) - C_{m1} m_1 \cos(\theta_1(t)) \sin(\theta_s(t)) - c_2 m_2 \cos(\theta_1(t)) \sin(\theta_s(t)) - c_3 m_2 \cos(\theta_s(t)) \sin(\theta_2(t)) - c_3 m_2 \cos(\theta_1(t)) \cos(\theta_2(t)) \sin(\theta_s(t))$$

$$C_1 = r(t)(M_c a_1 \cos(\theta_s(t)) + a_1 m_1 \cos(\theta_s(t)) + a_1 m_2 \cos(\theta_s(t)) + c_1 m_1 \cos(\theta_s(t)) + c_1 m_2 \cos(\theta_s(t)) + C_{m1} m_1 \cos(\theta_1(t)) \cos(\theta_s(t)) + c_2 m_2 \cos(\theta_1(t)) \cos(\theta_s(t)) - c_3 m_2 \sin(\theta_2(t)) \sin(\theta_s(t)) + c_3 m_2 \cos(\theta_1(t)) \cos(\theta_2(t)) \cos(\theta_s(t))) + M_c a_1^2 + a_1^2 m_1 + a_1^2 m_2 + c_1^2 m_1 + c_1^2 m_2 + c_3^2 m_2 + C_{m1}^2 m_1 \cos(\theta_1(t))^2 + c_2^2 m_2 \cos(\theta_1(t))^2 - c_3^2 m_2 \cos(\theta_2(t))^2 + 2a_1 c_1 m_1 + 2a_1 c_1 m_2 + 2C_{m1} a_1 m_1 \cos(\theta_1(t)) + 2C_{m1} c_1 m_1 \cos(\theta_1(t)) + 2a_1 c_2 m_2 \cos(\theta_1(t)) + 2c_1 c_2 m_2 \cos(\theta_1(t)) + c_3^2 m_2 \cos(\theta_1(t))^2 \cos(\theta_2(t))^2 + 2a_1 c_3 m_2 \cos(\theta_1(t)) \cos(\theta_2(t)) + 2c_1 c_3 m_2 \cos(\theta_1(t)) \cos(\theta_2(t)) + 2c_2 c_3 m_2 \cos(\theta_1(t))^2 \cos(\theta_2(t))$$

$$D_1 = c_2 m_2 \sin(\theta_1(t)) \sin(\theta_2(t)) c_3 - r(t)(C_{m1} m_1 \sin(\theta_1(t)) \sin(\theta_s(t)) + c_2 m_2 \sin(\theta_1(t)) \sin(\theta_s(t)) + c_3 m_2 \cos(\theta_2(t)) \sin(\theta_1(t)) \sin(\theta_s(t))) + m_2 \cos(\theta_2(t)) \sin(\theta_1(t)) \sin(\theta_2(t)) c_3^2$$

$$E_1 = r(t)(c_3 m_2 \cos(\theta_2(t)) \cos(\theta_s(t)) - c_3 m_2 \cos(\theta_1(t)) \sin(\theta_2(t)) \sin(\theta_s(t))) + c_3^2 m_2 \cos(\theta_1(t)) + a_1 c_3 m_2 \cos(\theta_2(t)) + c_1 c_3 m_2 \cos(\theta_2(t)) + c_2 c_3 m_2 \cos(\theta_1(t)) \cos(\theta_2(t))$$

……..

$$A_2 = -M_c a_1 \sin(\theta_s(t)) - a_1 m_1 \sin(\theta_s(t)) - a_1 m_2 \sin(\theta_s(t)) - c_1 m_1 \sin(\theta_s(t)) - c_1 m_2 \sin(\theta_s(t)) - C_{m1} m_1 \cos(\theta_1(t)) \sin(\theta_s(t)) - c_2 m_2 \cos(\theta_1(t)) \sin(\theta_s(t)) - c_3 m_2 \cos(\theta_s(t)) \sin(\theta_2(t)) - c_3 m_2 \cos(\theta_1(t)) \cos(\theta_2(t)) \sin(\theta_s(t))$$

$$. B_2 = M_c + m_1 + m_2$$

$$C_2 = -M_c a_1 \sin(\theta_s(t)) - a_1 m_1 \sin(\theta_s(t)) - a_1 m_2 \sin(\theta_s(t)) - c_1 m_1 \sin(\theta_s(t)) - c_1 m_2 \sin(\theta_s(t)) - C_{m1} m_1 \cos(\theta_1(t)) \sin(\theta_s(t)) - c_2 m_2 \cos(\theta_1(t)) \sin(\theta_s(t)) - c_3 m_2 \cos(\theta_s(t)) \sin(\theta_2(t)) - c_3 m_2 \cos(\theta_1(t)) \cos(\theta_2(t)) \sin(\theta_s(t))$$

$$D_2 = -C_{m1}m_1\cos(\theta_s(t))\sin(\theta_1(t)) - c_2m_2\cos(\theta_s(t))\sin(\theta_1(t)) - c_3m_2\cos(\theta_2(t))\cos(\theta_s(t))\sin(\theta_1(t))$$

$$E_2 = -c_3m_2\cos(\theta_2(t))\sin(\theta_s(t)) - c_3m_2\cos(\theta_1(t))\cos(\theta_s(t))\sin(\theta_2(t))$$

……

$$\begin{aligned}A_3 = r(t)(&M_ca_1\cos(\theta_s(t)) + a_1m_1\cos(\theta_s(t)) + a_1m_2\cos(\theta_s(t)) \\ &+ c_1m_1\cos(\theta_s(t)) + c_1m_2\cos(\theta_s(t)) \\ &+ C_{m1}m_1\cos(\theta_1(t))\cos(\theta_s(t)) + c_2m_2\cos(\theta_1(t))\cos(\theta_s(t)) \\ &- c_3m_2\sin(\theta_2(t))\sin(\theta_s(t)) \\ &+ c_3m_2\cos(\theta_1(t))\cos(\theta_2(t))\cos(\theta_s(t))) + M_ca_1^2 + a_1^2m_1 + a_1^2m_2 \\ &+ c_1^2m_1 + c_1^2m_2 + c_3^2m_2 + C_{m1}^2m_1\cos(\theta_1(t))^2 + c_2^2m_2\cos(\theta_1(t))^2 \\ &- c_3^2m_2\cos(\theta_2(t))^2 + 2a_1c_1m_1 + 2a_1c_1m_2 + 2C_{m1}a_1m_1\cos(\theta_1(t)) \\ &+ 2C_{m1}c_1m_1\cos(\theta_1(t)) + 2a_1c_2m_2\cos(\theta_1(t)) \\ &+ 2c_1c_2m_2\cos(\theta_1(t)) + c_3^2m_2\cos(\theta_1(t))^2\cos(\theta_2(t))^2 \\ &+ 2a_1c_3m_2\cos(\theta_1(t))\cos(\theta_2(t)) + 2c_1c_3m_2\cos(\theta_1(t))\cos(\theta_2(t)) \\ &+ 2c_2c_3m_2\cos(\theta_1(t))^2\cos(\theta_2(t))\end{aligned}$$

$$\begin{aligned}B_3 = -M_ca_1\sin(\theta_s(t)) - a_1m_1\sin(\theta_s(t)) - a_1m_2\sin(\theta_s(t)) - c_1m_1\sin(\theta_s(t)) \\ - c_1m_2\sin(\theta_s(t)) - C_{m1}m_1\cos(\theta_1(t))\sin(\theta_s(t)) \\ - c_2m_2\cos(\theta_1(t))\sin(\theta_s(t)) - c_3m_2\cos(\theta_s(t))\sin(\theta_2(t)) \\ - c_3m_2\cos(\theta_1(t))\cos(\theta_2(t))\sin(\theta_s(t))\end{aligned}$$

$$\begin{aligned}C_3 = I_c + M_ca_1^2 + a_1^2m_1 + a_1^2m_2 + c_1^2m_1 + c_1^2m_2 + c_3^2m_2 + C_{m1}^2m_1\cos(\theta_1(t))^2 \\ + c_2^2m_2\cos(\theta_1(t))^2 - c_3^2m_2\cos(\theta_2(t))^2 + 2a_1c_1m_1 + 2a_1c_1m_2 \\ + 2C_{m1}a_1m_1\cos(\theta_1(t)) + 2C_{m1}c_1m_1\cos(\theta_1(t)) \\ + 2a_1c_2m_2\cos(\theta_1(t)) + 2c_1c_2m_2\cos(\theta_1(t)) \\ + c_3^2m_2\cos(\theta_1(t))^2\cos(\theta_2(t))^2 + 2a_1c_3m_2\cos(\theta_1(t))\cos(\theta_2(t)) \\ + 2c_1c_3m_2\cos(\theta_1(t))\cos(\theta_2(t)) + 2c_2c_3m_2\cos(\theta_1(t))^2\cos(\theta_2(t))\end{aligned}$$

$$D_3 = m_2\cos(\theta_2(t))\sin(\theta_1(t))\sin(\theta_2(t))c_3^2 + c_2m_2\sin(\theta_1(t))\sin(\theta_2(t))c_3$$

$$\begin{aligned}E_3 = c_3^2m_2\cos(\theta_1(t)) + a_1c_3m_2\cos(\theta_2(t)) + c_1c_3m_2\cos(\theta_2(t)) \\ + c_2c_3m_2\cos(\theta_1(t))\cos(\theta_2(t))\end{aligned}$$

……

$$A_4 = c_2 m_2 \sin(\theta_1(t)) \sin(\theta_2(t)) c_3 - r(t)(C_{m1} m_1 \sin(\theta_1(t)) \sin(\theta_s(t)) + c_2 m_2 \sin(\theta_1(t)) \sin(\theta_s(t)) + c_3 m_2 \cos(\theta_2(t)) \sin(\theta_1(t)) \sin(\theta_s(t))) + m_2 \cos(\theta_2(t)) \sin(\theta_1(t)) \sin(\theta_2(t)) c_3^2$$

$$B_4 = -C_{m1} m_1 \cos(\theta_s(t)) \sin(\theta_1(t)) - c_2 m_2 \cos(\theta_s(t)) \sin(\theta_1(t)) - c_3 m_2 \cos(\theta_2(t)) \cos(\theta_s(t)) \sin(\theta_1(t))$$

$$C_4 = m_2 \cos(\theta_2(t)) \sin(\theta_1(t)) \sin(\theta_2(t)) c_3^2 + c_2 m_2 \sin(\theta_1(t)) \sin(\theta_2(t)) c_3$$

$$D_4 = m_1 C_{m1}^2 + m_2 c_2^2 + 2 m_2 c_2 c_3 \cos(\theta_2(t)) + m_2 c_3^2 \cos(\theta_2(t))^2$$

$$E_4 = 0$$

……

$$A_5 = r(t)(c_3 m_2 \cos(\theta_2(t)) \cos(\theta_s(t)) - c_3 m_2 \cos(\theta_1(t)) \sin(\theta_2(t)) \sin(\theta_s(t))) + c_3^2 m_2 \cos(\theta_1(t)) + a_1 c_3 m_2 \cos(\theta_2(t)) + c_1 c_3 m_2 \cos(\theta_2(t)) + c_2 c_3 m_2 \cos(\theta_1(t)) \cos(\theta_2(t))$$

$$B_5 = -c_3 m_2 \cos(\theta_2(t)) \sin(\theta_s(t)) - c_3 m_2 \cos(\theta_1(t)) \cos(\theta_s(t)) \sin(\theta_2(t))$$

$$C_5 = c_3^2 m_2 \cos(\theta_1(t)) + a_1 c_3 m_2 \cos(\theta_2(t)) + c_1 c_3 m_2 \cos(\theta_2(t)) + c_2 c_3 m_2 \cos(\theta_1(t)) \cos(\theta_2(t))$$

$$D_5 = 0$$

$$E_5 = c_3^2 m_2$$

Table 3 and table 4 show all the parameters that were used.

Chapter 5: Control System Design

This chapter endeavors to design a two-controller architecture for the system, leveraging linear control theories established within the academic community. The primary objective is to foster a deep understanding of the constraints inherent in linear control theories and explore potential strategies for mitigating these limitations.

5.1 Control of the cart

As observed in Chapter 4, the system's equations exhibit nonlinear behavior in both scenarios, whether analyzed independently or as a complete model. This nonlinearity presents a significant challenge for controller design. To address this, a novel model is developed, focusing solely on the cart's dynamics within a planar framework. The influence of the inclined plane is implicitly considered through the dynamics of the two motors, which can be neglected due to their relatively short settling times. Consequently, the primary objective shifts to determining the optimal force distribution required from each motor.

5.1.1 The dynamic model

A kinematic analysis of the system reveals three degrees of freedom: x, y, and θ. Here, x represents the linear displacement in the x-direction, y represents the linear displacement in the y-direction, and θ denotes the steering angle.

Based on the dynamic analysis, three equations are derived to describe the motion of the system. To simplify the analysis, frictional forces are neglected.

$$f_1 \cos\theta + f_2 \cos\theta = m\ddot{x} \tag{5.1}$$

$$\mathbf{f_1 \sin\theta + f_2 \sin\theta = m\ddot{y}} \tag{5.2}$$

$$f_1 \cdot L - f_2 \cdot L = I\ddot{\theta} \tag{5.3}$$

Table 13: Cart Parameter.

I (moment of inertia)	0.101 $kg \cdot m^2$
f_1 (force to be determined)	To be controlled
f_2(force to be determined)	To be controlled
L (distance from the wheel to CoM)	0.125 m
m (mass of the cart)	7 kg

As evident from (5.1), (5.2), and (5.3) the system exhibits nonlinear behavior. Furthermore, the coupling between these equations is mediated by a nonlinear input, rendering the model incompatible with conventional linear control theories. To address this challenge, the following solution is proposed:

$$f_1 + f_2 = m\ddot{r} \tag{5.4}$$

Where r is the total distance traveled by cart in either direction.

The control concept is as follows. Given $r^2(t) = x^2(t) + y^2(t)$ and the desired orientation $\theta_d = \tan^{-1}\left(\frac{y_d}{x_d}\right)$; these relationships enable the design of a linear controller. The desired orientation, θ_d, is used to generate an error signal, while *r(t)* is directly fed into the controller. This closed-loop architecture completes the feedback control system. A graphical representation of this control scheme is provided in the following Figure.

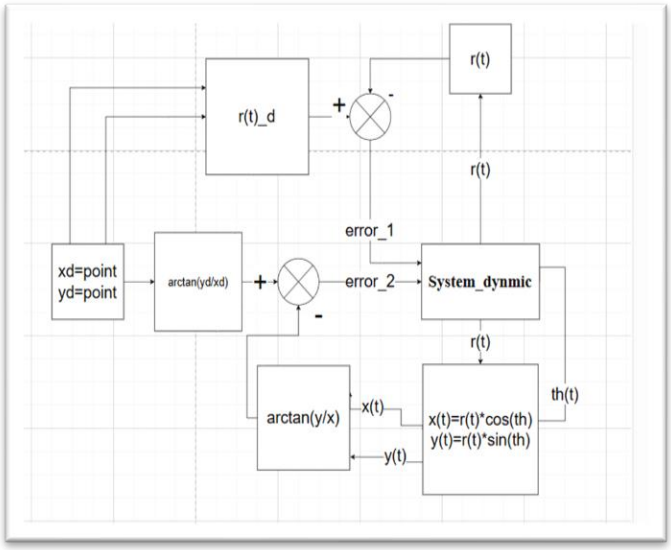


Figure 49: Control algorithm.

The state variables and equations would be described as following:

$$x_1 = r \qquad \dot{x}_1 = x_2$$

$$x_2 = \dot{r} \qquad \dot{x}_2 = \frac{1}{m}(f_1 + f_2)$$

$$x_3 = \theta \qquad \dot{x}_3 = x_4$$

$$x_4 = \dot{\theta} \qquad \dot{x}_4 = \frac{1}{I}(f_1 \cdot L - f_2 \cdot L)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \frac{1}{m} & \frac{1}{m} \\ 0 & 0 \\ \frac{1}{I} \cdot L & -\frac{1}{I} \cdot L \end{bmatrix} \begin{bmatrix} f1 \\ f2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

As observed in the previous state, matrix A exhibits rank deficiency, indicating singularity. While this might initially suggest an error, rank deficiency is a characteristic inherent to certain linear systems. The crucial determinant for proceeding with the analysis is the controllability matrix.

$$C_0 = \begin{bmatrix} 0 & 0 & 0.1429 & 0.1429 & 0 & 0 & 0 & 0 \\ 0.1429 & 0.1429 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1.25 & -1.25 & 0 & 0 & 0 & 0 \\ 1.25 & -1.25 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Where C_0 is the controllability matrix, $Rank(c_0) \geq n$

- n=4, Then the system is fully controllable.

Given the computational complexity of the problem, the remaining calculations will be performed using MATLAB.

Control law:

$$\boldsymbol{u = \frac{k_i}{s}(R - y) - kx} \tag{5.5}$$

Where:

- k_i is the Integral gain.
- R is Reference input to track.
- k is State feedback gain.

The system under consideration involves a motor-driven mechanism with a gear wheel. The motor generates a torque, which is linearly related to the force vector 'u', a 2x1 matrix representing the force applied to the system. Assuming negligible motor dynamics and considering the combined inertia of the motor and gear wheel, the system dynamics can be analyzed.

The vector 'u' represents the control signal generated by a PI controller. It is important to note that the concept of PI control is inherently associated with single-input single-output (SISO) systems, specifically referring to the control law where the PI effect is achieved.

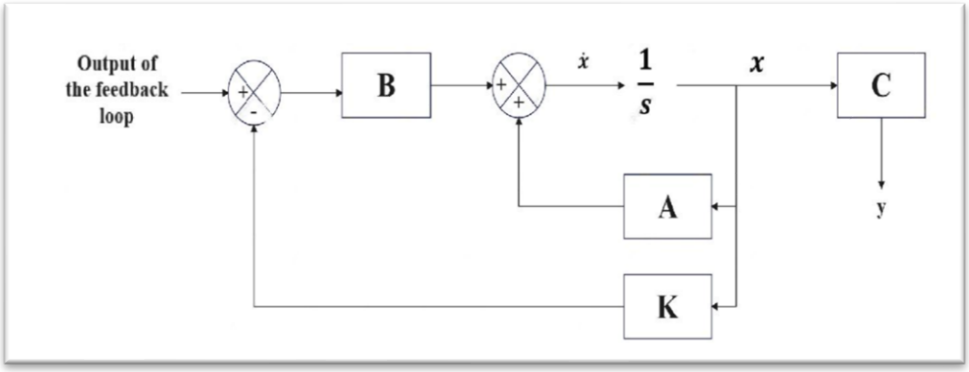


Figure 51: Inner Loop.

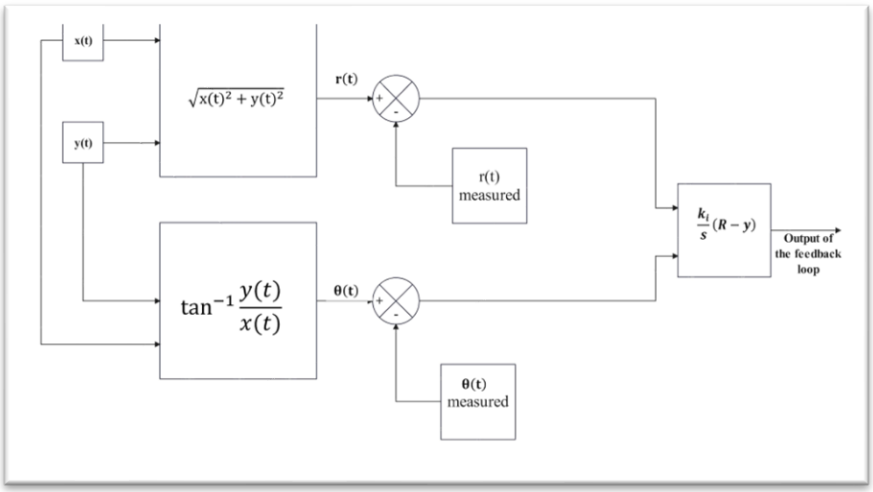


Figure 50: Outer Loop.

Given the computational complexity of the problem, the remaining calculations will be performed using MATLAB.

5.1.2 Simulation

First trial:

Table 14: Conditions of First trail.

Initial condition	[9 0 0 0 0 0]
Point /trajectory	$\begin{bmatrix} x(t) = 9cos\,t \\ y(t) = 9\,sin\,t \end{bmatrix}$

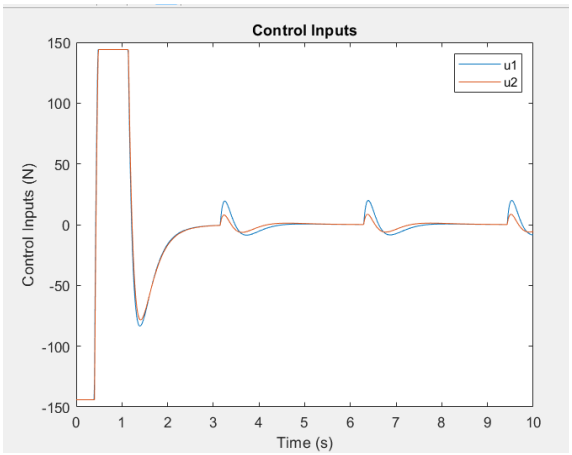


Figure 52: (First trail) Control inputs.

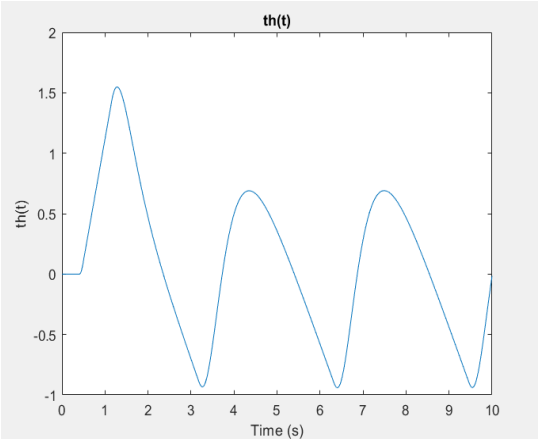


Figure 53: (First trail) th(t) over time.

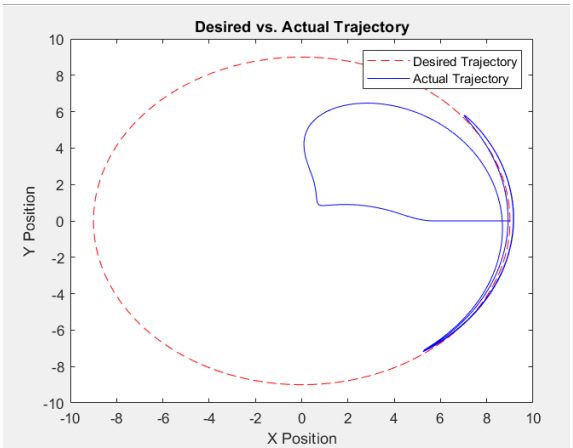


Figure 54: (First trail) Desired vs. Actual Trajectory.

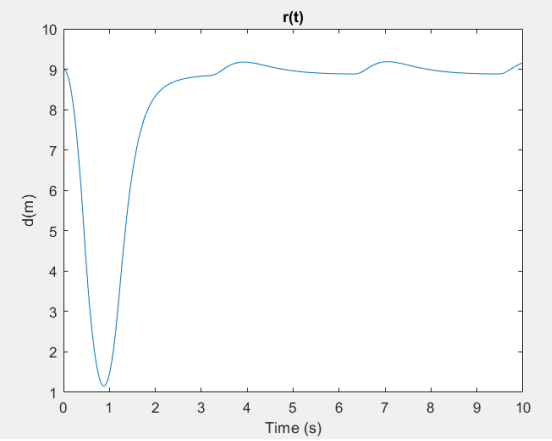


Figure 55: (First trail) r(t) over time.

As illustrated in Figure 52, Figure 53,Figure 54 and Figure 55,a steady-state error persists despite the controller design's objective of eliminating it. While the controller exhibits satisfactory regulation, it is crucial to acknowledge that the MATLAB simulations provide a virtual representation of the system's behavior. Consequently, extending the simulation time may reveal more pronounced steady-state errors. A brief analysis of each figure follows.

- **Figure 52** reveals a discrepancy between the expected and observed steady-state values of the input. This deviation can be attributed to the inherent nonlinearity of the path trajectory itself.
- **Figure 53** demonstrates that the angle stabilizes, converging to a finite value. This is a positive observation, indicating that the system does not exhibit unbounded growth in this parameter.
- **Figure 54** demonstrates that the system exhibits some tracking behavior, converging towards the desired trajectory while maintaining a bounded error envelope. Notably, this tracking error persists even when the system's initial condition is located directly on the desired trajectory. To address this persistent tracking error and potentially enhance tracking accuracy, the implementation of a nonlinear control design incorporating a reaching law could be a viable solution.
- **Figure 55** demonstrates that the system converges to a finite value.

Second trial:

Table 15: Conditions of Second trail.

Initial condition	[0 0 0 0 0 0]
Point /trajectory	$\begin{bmatrix} x(t) = 9 \\ y(t) = 3 \end{bmatrix}$

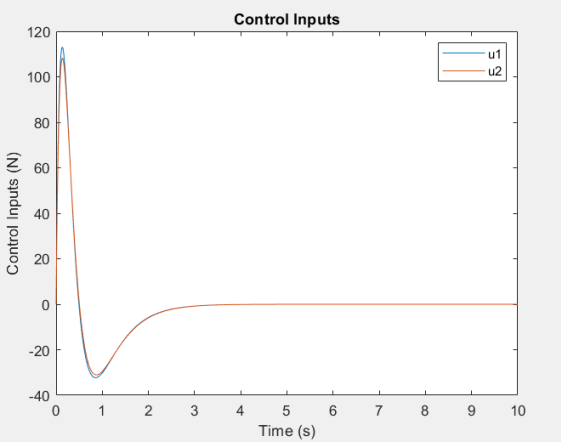


Figure 56: (Second trail) Control inputs.

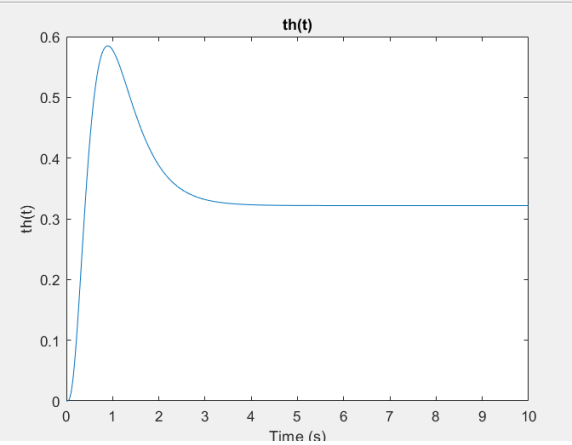


Figure 57: (First trail) th(t) over time.

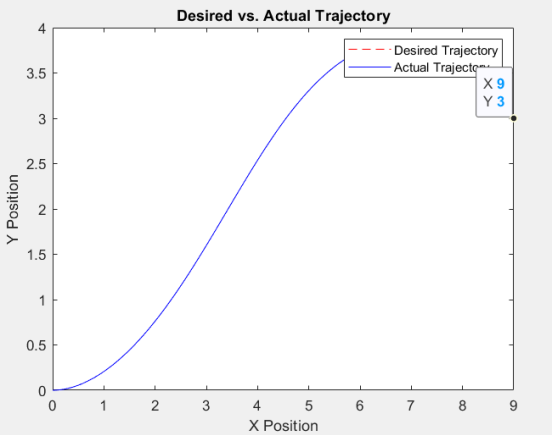


Figure 58: (Second trail) Desired vs. Actual Trajectory.

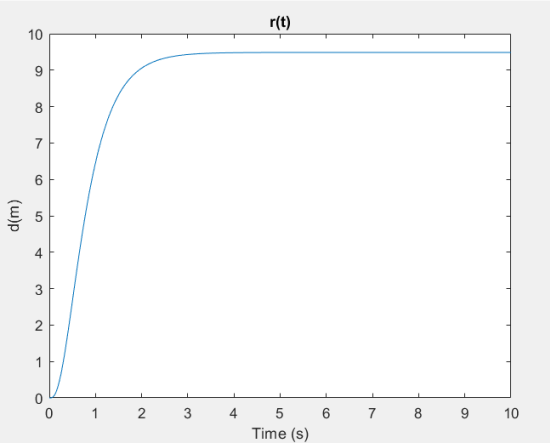


Figure 59: (Second trail) r(t) over time.

- **Figure 56** indicates that the input signal converges to a steady-state value. This observation suggests that the system may exhibit zero steady-state error.
- **Figure 57** also indicates that the angle converges to a steady-state value.
- **Figure 58** indicates that the system exhibits zero steady-state error.
- **Figure 59** also demonstrates that the system converges to a steady-state value.

5.2 Control of the arm

For the manipulator under consideration, the development of a new dynamic model is unnecessary. The dynamic model derived in Chapter 4 will be utilized for control purposes. However, to facilitate the implementation of the PID controller, linearization of the model is required. To this end, the moment of inertia matrix, represented by M, will be evaluated at its maximum values. Subsequently, the controller will be designed while neglecting other terms, including gravity.

5.2.1 Dynamic model used in the design

$$\tau_1 = I_1 \ddot{\theta}_1 \tag{5.5}$$

$$\tau_2 = I_2 \ddot{\theta}_2 \tag{5.6}$$

$$I_1 = m_1 d_1^2 + m_2 d_1^2 + 2 m_2 d_2 d_l + m_2 d_2^2 \tag{5.7}$$

$$I_2 = m_2 d_2^2 \tag{5.8}$$

Table 16: Parameters of the manipualtor.

Parmeter	Description	Value
θ_1	Servo y	To be controlled
θ_2	Servo x	To be controlled
d_1	Length of the first link	10 cm
d_2	Length of the second link	10 cm
m_1	Mass of the first link	100 g
m_2	Mass of the second link	350 g

Furthermore, it is important to note that the inertia associated with the object's dimensions is neglected in this analysis, as the system is modeled as a point mass.

The control design will be implemented in Simulink. Given that the approximated model exhibits minimal coupling, the system can be effectively represented as two independent subsystems within the Simulink environment.

5.2.2 simulation

Open loop:

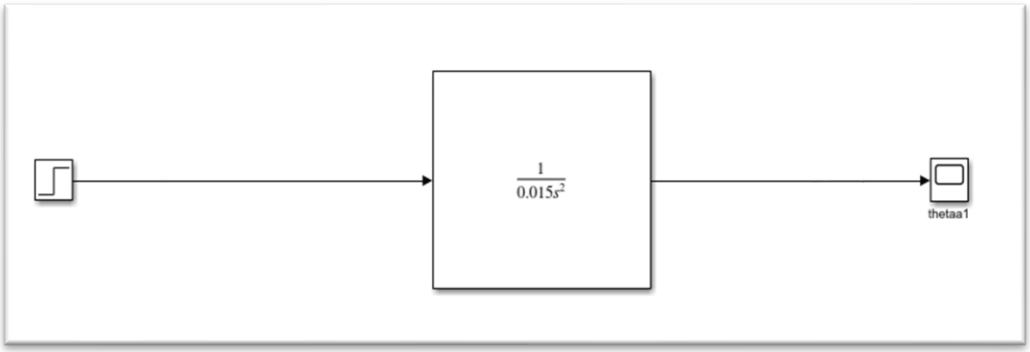


Figure 60 : Open loop block for the first link.

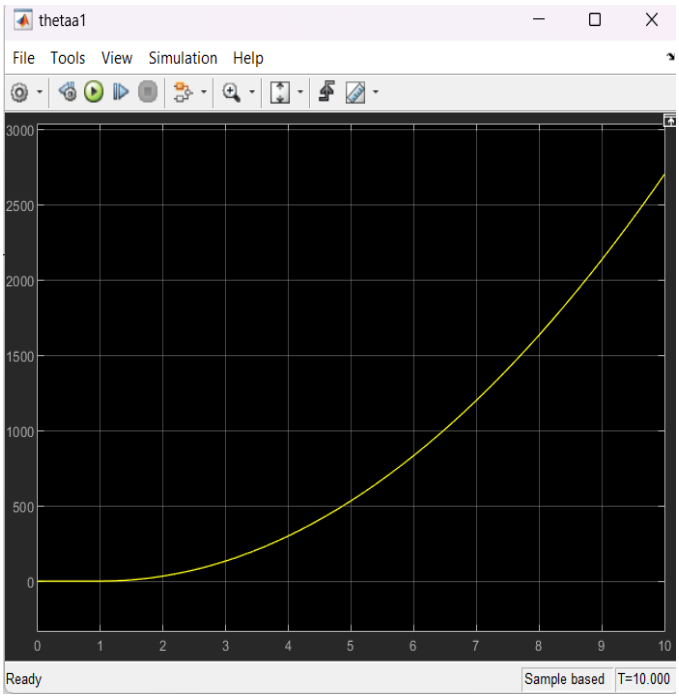


Figure 61: Open loop block for the first link simulation.

As observed in Figure 61 the system exhibits unstable behavior.

Closed loop PID:

$$G_C = kp + \frac{k_i}{s} + k_d s \tag{5.9}$$

Equation (5.9) represents the transfer function of the PID controller.

Several methods exist for tuning PID controllers, including trial and error, Ziegler-Nichols, root locus analysis, and MATLAB's built-in tuning tools. In this project, the PID controller parameters were determined using the MATLAB Tuner app.

Table 17: Specifications of the simulation.

Input to be track	$\frac{\pi}{4}$
Controlled variable	θ_1

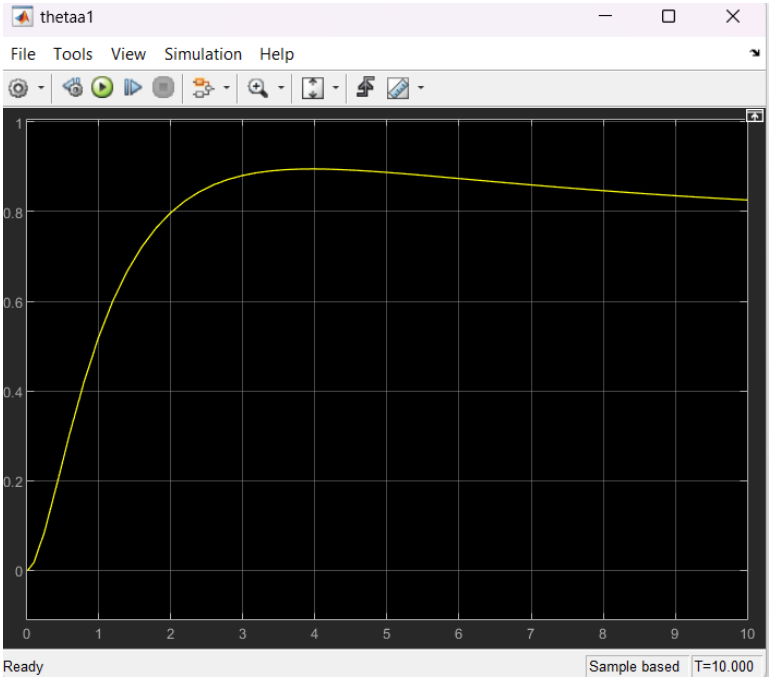


Figure 62: Closed loop simulation for the first link.

As illustrated in Figure 60, the system converges to a steady-state value.

Chapter 6: Conclusions and Recommendations

The development of semi-autonomous mobile robot cleaners represents a significant leap forward in solar panels maintenance. These innovative devices offer a range of benefits and capabilities that enhance efficiency and effectiveness across various sectors.

6.1 Conclusions

In conclusion, the design and development of PanelGaurd, a semi-autonomous solar panel cleaning robot, presents a promising solution to the challenge of maintaining solar panel efficiency. By automating the cleaning process, PanelGaurd not only reduces the need for manual labor and associated risks but also ensures consistent and thorough cleaning, leading to improved energy production. The integration of various sensors, a robust control system, and a durable mechanical structure further enhances the robot's effectiveness and adaptability to different environments.

6.2 Results

The results demonstrate that the robot performs its intended task effectively. However, several conclusions have been drawn from the findings:

- The robot's chassis, tracked wheels, and complete mechanical assembly were designed and constructed entirely from scratch, and this was accomplished successfully and with precision.

- It was found that the overall cost of the robot as a product could be considered reasonable and suitable for its intended purpose.

- It was planned to integrate a water tank into the robot. However, after conducting research and visiting solar farms, it was observed that cleaning solar panels requires a significantly large amount of water. Consequently, it was decided to utilize an external water source instead.

- Safety: The robot has been successfully designed to operate safely, ensuring it does not cause any harm to users or damage to the solar panels.

- The robot is designed to accommodate various solar panel installations, considering variations in height and inclination.

- Environmental Impact: Using a robot ensures that the cleaning process is environmentally friendly. It contains no chemical by-products and promotes sustainable energy production.

- Low Maintenance Requirements: The robot is designed to require minimal maintenance, which reduces downtime and keeps operational costs low overtime.

6.3 Recommendations

PanelGuard was designed as a prototype; however, if it is to be developed into a commercial product, several recommendations should be considered:

1- Integrating a camera into PanelGuard:

Integrate a camera system into PanelGuard to provide real-time visual feedback during the cleaning process. This will help users monitor the cleaning progress and ensure the robot is functioning properly. The camera could also assist in detecting dirt levels, obstacles, or panel damage, allowing the robot to adjust its cleaning strategy automatically for more efficient operation. Additionally, the camera can support remote monitoring, allowing users to oversee operations from a distance.

It should be noted that an additional controller must be integrated if a camera is added.

2- Enhanced Durability and Reliability:

- Ensure that all components of the robot are built to last and can handle different environmental conditions, including extreme temperatures, high humidity, dust, and UV exposure.
- Test the robot to make sure it works reliably over time without frequent breakdowns.

3- Improved Power Efficiency:

- Optimize the robot's power consumption to increase operational time and reduce the need for frequent recharging.
- Allow the robot to recharge itself by drawing power from the panel itself to make the robot self-sufficient in terms of power.

4- Advanced Sensors and AI Integration:

- Integrate more advanced sensors for detecting dirt levels and automatically adjusting the cleaning strategy based on the panel's condition.
- Incorporate artificial intelligence for optimizing cleaning paths, identifying problem areas, and adapting to new panel configurations or obstacles.

5- Thermal Management System:

- Incorporate a thermal management system to prevent overheating of the robot's components, especially in regions with high temperatures or under continuous use. This could involve passive cooling techniques or temperature sensors.

Chapter 7: References

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Appendices:

Appendix A: ESP32 code

```
#include <ESP32Servo.h>

#include <Bluepad32.h>

#include <Arduino.h>

ControllerPtr myControllers[BP32_MAX_GAMEPADS];

// PIN CONNECTIONS

#define LED_PIN 2

int ENApin = 5;  // motor 1 speed

int IN1pin = 18; // motor 1 dir1

int IN2pin = 19; // motor 1 dir2

int ENBpin = 27; // motor 2 speed

int IN3pin = 26; // motor 2 dir1

int IN4pin = 25; // motor 2 dir2

int xServoPin = 12; // sweeping servo

int yServoPin = 14; // lifting servo

int IN5pin = 32; // Brush motor dir1

int IN6pin = 33; // Brush motor dir2
```

```
Servo xServo;

Servo yServo;

int sweep = 0;

int B = 0;

int y = 0;

int up = 0;

int down = 0;

int right = 0;

int left = 0;

int pos = 122;

int speed1 = 0;

int speed2 = 0;

int start = 0;

void onConnectedController(ControllerPtr ctl) {

 bool foundEmptySlot = false;

 for (int i = 0; i < BP32_MAX_GAMEPADS; i++) {

  if (myControllers[i] == nullptr) {

    Serial.printf("CALLBACK: Controller is connected, index=%d\n", i);

    ControllerProperties properties = ctl->getProperties();
```

```
      Serial.printf("Controller model: %s, VID=0x%04x, PID=0x%04x\n", ctl->getModelName().c_str(), properties.vendor_id, properties.product_id);
      myControllers[i] = ctl;
      foundEmptySlot = true;
      break;
    }
  }
  if (!foundEmptySlot) {
    Serial.println("CALLBACK: Controller connected, but could not found empty slot");
  }
}
void onDisconnectedController(ControllerPtr ctl) {
  bool foundController = false;
  for (int i = 0; i < BP32_MAX_GAMEPADS; i++) {
    if (myControllers[i] == ctl) {
      Serial.printf("CALLBACK: Controller disconnected from index=%d\n", i);
      myControllers[i] = nullptr;
      foundController = true;
      break;
```

```
    }
  }
  if (!foundController) {
    Serial.println("CALLBACK: Controller disconnected, but not found in myControllers");
  }
}
void handleButtonPress(ControllerPtr ctl, int button, int &counter) {
  if (ctl->buttons() == button) {
    counter++;
    delay(10);
  }
}
void StopMotors() {
  digitalWrite(IN1pin, LOW);
  digitalWrite(IN2pin, LOW);
  analogWrite(ENApin, 0);
  digitalWrite(IN3pin, LOW);
  digitalWrite(IN4pin, LOW);
  analogWrite(ENBpin, 0);
```

```
}

void moveForward() {

    digitalWrite(IN1pin, HIGH);

    digitalWrite(IN2pin, LOW);

    analogWrite(ENApin, 150);

    digitalWrite(IN3pin, HIGH);

    digitalWrite(IN4pin, LOW);

    analogWrite(ENBpin, 158);

}

void moveBackward() {

    digitalWrite(IN1pin, LOW);

    digitalWrite(IN2pin, HIGH);

    analogWrite(ENApin, 150);

    digitalWrite(IN3pin, LOW);

    digitalWrite(IN4pin, HIGH);

    analogWrite(ENBpin, 158);

}

void turnLeft() {

    digitalWrite(IN1pin, LOW);

    digitalWrite(IN2pin, HIGH);
```

```
    analogWrite(ENApin, 130);
    digitalWrite(IN3pin, HIGH);
    digitalWrite(IN4pin, LOW);
    analogWrite(ENBpin, 130);
}
void turnRight() {
    digitalWrite(IN1pin, HIGH);
    digitalWrite(IN2pin, LOW);
    analogWrite(ENApin, 130);
    digitalWrite(IN3pin, LOW);
    digitalWrite(IN4pin, HIGH);
    analogWrite(ENBpin, 130);
}
void processGamepad(ControllerPtr ctl) {
  handleButtonPress(ctl, 0x0001, start);
  if (start % 2 != 0 && start != 0) {
    digitalWrite(LED_PIN, HIGH);
    handleButtonPress(ctl, 0x0080, up);
    handleButtonPress(ctl, 0x0040, down);
    handleButtonPress(ctl, 0x0010, right);
```

```
  handleButtonPress(ctl, 0x0020, left);
  if (digitalRead(2) == HIGH) {
    if (up % 2 != 0 && up != 0) {
      moveForward();
    } else if (down % 2 != 0 && down != 0) {
      moveBackward();
    } else if (left % 2 != 0 && left != 0) {
      turnLeft();
    } else if (right % 2 != 0 && right != 0) {
      turnRight();
    } else {
      StopMotors();
    }
  }
} else if (start % 2 == 0) {
  digitalWrite(LED_PIN, LOW);
  up = 2;
  left = 2;
  right = 2;
  down = 2;
```

```
  StopMotors();

}

handleButtonPress(ctl, 0x0004, y);

handleButtonPress(ctl, 0x0008, B);

if (B % 2 != 0 && y % 2 != 0 && B != 0 && y != 0) {

  delay(50);

  digitalWrite(IN5pin, HIGH);

  digitalWrite(IN6pin, LOW);

} else {

  digitalWrite(IN5pin, LOW);

  digitalWrite(IN6pin, LOW);

}

if (y % 2 != 0 && y != 0) {

  delay(100);

  yServo.write(0);

   delay(600);

} else {

  delay(100);

  xServo.write(122);

  delay(400);
```

```
  digitalWrite(IN5pin, LOW);
  digitalWrite(IN6pin, LOW);
  delay(600);
  yServo.write(70);
 }
if (y % 2 != 0 && y != 0) {
  if (ctl->axisRX() >= 300) {
    sweep = 0;
    delay(50);
    sweepServo(xServo.read(), 67, 5);
  } else if (ctl->axisRX() <= -300) {
    sweep = 0;
    delay(50);
    sweepServo(xServo.read(), 177, 5);
  }
  handleButtonPress(ctl, 0x0002, sweep);
  if (sweep % 2 != 0 && sweep != 0) {
    sweepServo(67, 77, 10);
    sweepServo(77, 167, 5);
    sweepServo(167, 177, 10);
```

```
      sweepServo(177, 167, 10);

      sweepServo(167, 77, 5);

      sweepServo(77, 67, 10);

    } else if (sweep % 2 == 0) {

      xServo.write(122);

    }

  } else {

    xServo.write(122);

  }

}

void sweepServo(int startPos, int endPos, int delayTime) {

  for (int pos = startPos; (startPos < endPos) ? pos <= endPos : pos >= endPos;
(startPos < endPos) ? pos++ : pos--) {

    xServo.write(pos);

    delay(delayTime);

  }

}

void processControllers() {

  for (auto myController : myControllers) {
```

```
    if (myController && myController->isConnected() && myController->hasData()) {
      if (myController->isGamepad()) {
        processGamepad(myController);
      } else {
        Serial.println("Unsupported controller");
      }
    }
  }
}
void setup() {
  pinMode(LED_PIN, OUTPUT);
  pinMode(ENApin, OUTPUT);
  pinMode(IN1pin, OUTPUT);
  pinMode(IN2pin, OUTPUT);
  pinMode(ENBpin, OUTPUT);
  pinMode(IN3pin, OUTPUT);
  pinMode(IN4pin, OUTPUT);
  pinMode(IN5pin, OUTPUT);
  pinMode(IN6pin, OUTPUT);
```

```
  xServo.attach(xServoPin);

  yServo.attach(yServoPin);

  yServo.write(140);

  xServo.write(122);

    Serial.begin(115200);

  Serial.printf("Firmware: %s\n", BP32.firmwareVersion());

  const uint8_t* addr = BP32.localBdAddress();

  Serial.printf("BD Addr: %2X:%2X:%2X:%2X:%2X:%2X\n", addr[0], addr[1],
addr[2], addr[3], addr[4], addr[5]);

  BP32.setup(&onConnectedController, &onDisconnectedController);

  BP32.forgetBluetoothKeys();

  BP32.enableVirtualDevice(false);

}

void loop() {

  bool dataUpdated = BP32.update();

  if (dataUpdated)

  processControllers();

  delay(15);

}
```

Appendix B: Design of a full state feedback controller

```
% Define system matrices

A = [0 1 0 0; 0 0 0 0; 0 0 0 1; 0 0 0 0];

B = [0 0; 1/7 1/7; 0 0; (1/0.1)*0.125 -(1/0.1)*0.125];

C = [1 0 0 0; 0 0 1 0];

% Augment system matrices

A_aug = [A, zeros(size(A, 1), size(C, 1)); -C, zeros(size(C, 1))];

B_aug = [B; zeros(size(C, 1), size(B, 2))];

% Desired poles for the augmented system

desired_poles = [-2, -3, -4, -5, -6, -7];

% Place poles to get feedback gains

K_aug = place(A_aug, B_aug, desired_poles);

K = K_aug(:, 1:size(A, 2));

Ki = K_aug(:, size(A, 2) + 1:end);

% Display gains

disp('State feedback gain K:');

disp(K);

disp('Integral gain Ki:');
```

```
disp(Ki);

% Simulation parameters

T = 10; % Simulation time

dt = 0.01; % Time step

time = 0:dt:T;

% Initialize state and integral states

x = zeros(size(A, 1), 1);

zeta = zeros(size(C, 1), 1);

x0 = [0; 0; 0; 0]; % Initial conditions for the states

zeta0 = [0; 0]; % Initial conditions for the integral states

x = x0;

zeta = zeta0;

% Initialize arrays to store results

X = zeros(length(x), length(time));

Zeta = zeros(length(zeta), length(time));

U = zeros(size(B, 2), length(time));

% Simulation loop

u_max = 144; % Set the maximum limit for control inputs

for i = 1:length(time)

    % Desired Cartesian coordinates
```

```
    x_cartesian(i) = 9;

    y_cartesian(i) = 3;

    r1 = sqrt(x_cartesian(i)^2 + y_cartesian(i)^2);

    r2 = atan(y_cartesian(i) / x_cartesian(i));

    r = [r1; r2];

    % Compute control input

    u = -K * x - Ki * zeta;

    % Apply saturation limits to the control inputs

    u = max(min(u, u_max), -u_max);

    % Store results

    X(:, i) = x;

    Zeta(:, i) = zeta;

    U(:, i) = u;

    % Update states

    dx = A * x + B * u;

    dzeta = r - C * x;

    x = x + dx * dt;

    zeta = zeta + dzeta * dt;

end

% Actual Cartesian coordinates
```

```
x_actual = X(1, :) .* cos(X(3, :));

y_actual = X(1, :) .* sin(X(3, :));

% Plot the results

figure;

plot(time, X(1,:));

title('r(t)');

xlabel('Time (s)');

ylabel('d(m)');

figure;

plot(time, Zeta);

title('Integral States');

xlabel('Time (s)');

ylabel('Integral States');

legend('zeta1', 'zeta2');

figure;

plot(time, U);

title('Control Inputs');

xlabel('Time (s)');

ylabel('Control Inputs (N)');

legend('u1', 'u2');
```

```
% Plot desired and actual trajectories
figure;
plot(x_cartesian, y_cartesian, 'r--', 'DisplayName', 'Desired Trajectory');
hold on;
plot(x_actual, y_actual, 'b', 'DisplayName', 'Actual Trajectory');
xlabel('X Position');
ylabel('Y Position');
title('Desired vs. Actual Trajectory');
legend('show');
hold off;
figure;
plot(time, X(3,:));
title('th(t)');
xlabel('Time (s)');
ylabel('th(t)');
```